



APPENDIX OUTLINE

INTRODUCTION

WHY STUDY NAIL CARE

DESCRIBE THE NAIL UNIT

RECOGNIZE NAIL DISORDERS AND DISEASES

LEARN ABOUT MANICURING

PROCEDURES A1 TO A4

LEARNING OBJECTIVES

After reading through the Appendix, you will be able to:

LO¹

Describe the composition of the nail.

LO²

Identify common nail disorders and diseases.

LO³

Identify nail care equipment, implements, materials, and products.

LO⁴

Recognize the five general shapes of nails.

LO⁵

Understand table setup, hand massage, and manicure procedures.

Introduction

Barbershops of the first half of the twentieth century routinely provided manicures as part of the traditional shave, haircut, and shoeshine service. Today's barbershops can offer the service through a licensed manicurist, nail technician, or in a few states, by a licensed barber. Always refer to your state board rules and regulations to confirm the services you are licensed to perform.

why study NAIL CARE?

Barbers should study and have a thorough understanding of nail care because:

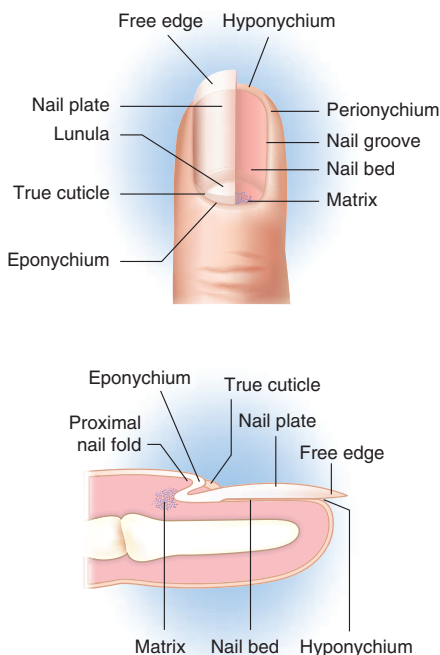
- They need to know the procedures for performing a manicure or to oversee a manicure that is performed by others.
- They need to recognize conditions that may benefit from a manicure service.
- They need to recognize conditions that prohibit a manicure service.

Describe the Nail Unit

After reading this section, you will be able to:

LO 1 Describe the composition of the nail.

figure A-1
Structure of the natural nail.



The **nail** (NAYL) is a horny, translucent plate of hard keratin that serves to protect the tips of the fingers and toes. Nails are part of the integumentary system and are considered to be appendages of the skin. The technical term for nail is **onyx** (AHN-iks).

The condition of the nail, like that of the skin, reflects the general health of the body. The normal, healthy nail is firm, flexible, and translucent with the pinkish color of the nail bed below showing through. Its surface should be smooth, curved, and unspotted, without any hollows or wavy **ridges** (RIJ-ez). No nerves or blood vessels are contained within the horny nail plate. The nail unit consists of the nail plate, nail bed, matrix, cuticle, eponychium, perionychium, hyponychium, specialized ligaments, and nail folds (**Figure A-1**).

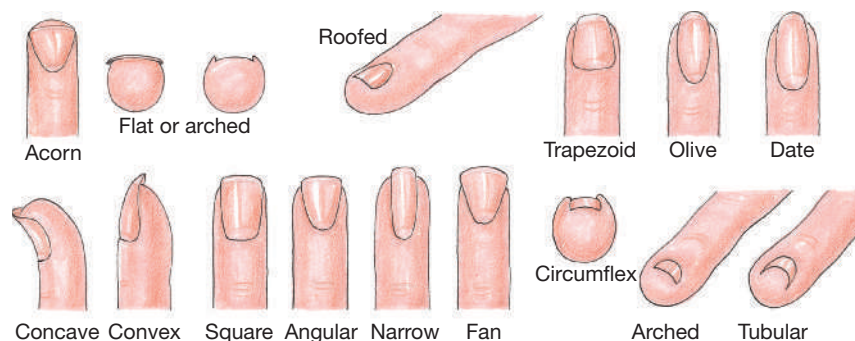
- The **nail plate** (NAYL PLAYT) is a hardened keratin plate, constructed of layers of matrix cells, that sits on and covers the nail bed. The nail plate slowly slides upon the nail bed as it grows and extends to the **free edge** (FREE EJ) of the nail.
- The **nail bed** (NAYL BED) is the portion of living skin that supports the nail plate as it grows toward the free edge. It is supplied with blood vessels that provide the pinkish tone from the lunula almost to the free edge and nerves that are attached to the nail plate. The nail bed is attached to the nail plate by a thin layer of tissue called the **bed epithelium** (BED ep-ih-TH-EE-lee-um), which helps guide the nail plate along the nail bed as it grows.

- The **matrix** (MAY-triks) is where the nail plate cells are formed. The cells are nourished by nerves, lymph, and blood vessels, and continue to grow as long as the matrix receives nutrition and remains healthy. The visible portion of the matrix is called the **lunula** (LOO-nuh-luh) or half-moon. It is located at the base of the nail where the matrix and the connective tissue of the nail bed join.
- The **cuticle** (KYOO-tih-kul) is the crescent of *dead*, colorless tissue attached to the nail plate around the base of the nail that forms a seal between the eponychium and the nail plate to help prevent injury and infection.
- The **eponychium** (ep-oh-NIK-ee-um) is the *living* skin at the base of the nail plate covering the matrix area and should not be confused with the cuticle.
- The **perionychium** (payr-ee-uh-NIK-ee-um) is the living skin bordering the root and sides of a nail.
- The **hyponychium** (hy-poh-NIK-ee-um) is the slightly thickened layer of skin between the fingertip and the free edge of the nail plate that forms a protective barrier against microorganisms and infection of the nail bed.
- The **specialized ligaments** form bands of fibrous tissue that attach the nail bed and matrix to the underlying bone that are located at the base of the matrix and around the edges of the nail bed.
- The **nail folds** (NAYL FOHLDZ) are folds of normal skin that surround the nail plate. These folds form the **nail grooves** (NAYL GROOVZ) on the sides of the nail that permit the nail to move as it grows.

NAIL GROWTH

Nail growth is influenced by nutrition, health, and disease. A normal, healthy nail grows forward, starting at the matrix and extending over the fingertip. The average rate of growth in the normal adult is about $\frac{1}{10}$ inch per month. Typically, nails grow faster in summer than they do in winter. Children's nails grow more rapidly, whereas those of elderly persons grow more slowly. The nail of the middle finger grows the fastest and the thumb-nail the most slowly. Although toenails grow more slowly than fingernails, they are thicker and harder (Figure A-2).

figure A-2
Various shapes of nails.



NAIL MALFORMATION

If the nail is separated from the nail bed through injury, it becomes distorted or **discolored** (dis-KUL-urd). Should the nail bed be injured after the loss of a nail, a badly formed new nail will result.

Nails are not shed in the same way that hair is shed. If a nail is torn off accidentally or lost through infection or disease, it will be replaced only if the matrix remains in good condition. In some cases, the replacement nails are shaped abnormally, due to interference at the base of the nail. Replacement of the nail takes about four to six months.

Recognize Nail Disorders and Diseases

After reading this section, you will be able to:

LO2 Identify common nail disorders and diseases.

A *nail disorder* is a condition caused by injury to the nail, disease, or a chemical or nutritional imbalance. Most clients will have experienced a common nail disorder at some time in their lives. If a client has a nail disorder, you may be able to help them in one of two ways: either cosmetically improve the condition or appearance of the nail plate or refer them to a physician. Be observant and if the nail or the skin surrounding it looks infected or inflamed, do not perform the manicure and recommend the client seek medical attention. Review **Tables A-1** and **A-2** to be able to recognize common nail disorders.

table A-1
OVERVIEW OF NAIL DISORDERS

Disorder	Signs or Symptoms
Beau’s Lines (BOWZ LYNZ)	Visible depressions running across the width of the natural nail plate; usually a result of major illness or injury that has traumatized the body.
Blue Fingernails	Blue or purple nail bed, usually from poor circulation.
Bruised Nail Beds	Dark purplish spots, usually due to physical injury.
Discolored Nails	Nails turn a variety of colors; may indicate surface staining, a systemic disorder, or poor blood circulation.
Eggshell Nails	Noticeably thin, white nail plates that are more flexible than normal and can curve over the free edge; usually caused by improper diet, hereditary factors, internal disease, or medication.
Hangnails (HANG-NAYLZ)	Living skin around the nail plate (often the eponychium) that becomes split or torn.
Koilonychia (koyal-oh-NICK-ee-uh)	Also known as <i>spoon nails</i> ; inverted or concave nails.

Leukonychia Spots (loo-koh-NIK-ee-ah SPATZ)	Also known as <i>white spots</i> ; whitish discolorations of the nail usually caused by minor injury to the nail matrix. Not related to the body's health or vitamin deficiencies.
Melanonychia (mel-uh-nuh-NIK-ee-uh)	Darkening of the fingernails or toenails; may be seen as a black band within the nail plate extending from the base to the free edge.
Nail Pterygium (NAYL teh-RIJ-ee-um)	Abnormal stretching of skin around the nail plate; usually caused by serious injury, such as burns, or an adverse skin reaction to chemical nail enhancement products or an allergic skin reaction.
Onychauxis (ahn-ih-KAHK-sis)	Thickening of the fingernails or toenails.
Onychogryposis (ahn-ih-koh-gry-POH-sis)	Also known as <i>ram's horn</i> or <i>claw nails</i> ; an enlargement of the fingernails or toenails accompanied by increased thickening and curvature.
Onychophagy (ahn-ih-koh-FAY-jee)	Also known as <i>bitten nails</i> ; chewed nails or chewed hardened skin surrounding the nail plate.
Onychorrhexis (ahn-ih-koh-REK-sis)	Split or brittle nails that have a series of lengthwise ridges giving a rough appearance to the surface of the nail plate.
Pincer Nail (PIN-sir NAYL)	Also known as <i>trumpet nail</i> ; increased crosswise curvature throughout the nail plate caused by an increased curvature of the matrix; the edges of the nail plate may curl around to form the shape of a trumpet or sharp cone at the free edge.
Plicatured Nail (plik-a-CHOORD NAYL)	Also known as <i>folded nail</i> ; a type of highly curved nail plate, usually caused by injury to the matrix, but it may be inherited.
Ridges	Vertical lines running the length of the natural nail plate that are caused by uneven growth of the nails, usually the result of normal aging.
Splinter Hemorrhage (SPLIN-tohr HEM-err-aje)	Physical trauma or injury to the nail bed that damages the capillaries and allows a small amount of blood flow.

table A-2

OVERVIEW OF NAIL DISEASES

Disease	Signs or Symptoms
Nail Psoriasis (NAYL suh-RY-uh-sis)	Tiny pits or severe roughness on the surface of the nail plate.
Onychia (uh-NIK-ee-uh)	Inflammation of the nail matrix followed by shedding of the nail.
Onychocryptosis (ahn-ih-koh-krip-TOH-sis)	Also known as <i>ingrown nails</i> ; nail grows into the sides of the tissue around the nail.
Onycholysis (ahn-ih-KAHL-ih-sis)	Lifting of the nail plate from the nail bed, without shedding, usually beginning at the free edge and continuing toward the lunula area.
Onychomadesis (ahn-ih-koh-muh-DEE-sis)	Separation and falling off of a nail plate from the nail bed; can affect fingernails and toenails.

(Continues)

table A-2 (Continued)

Onychomycosis (ahn-ih-koh-my-KOH-sis)	Fungal infection of the natural nail plate.
Paronychia (payr-uh-NIK-ee-uh)	Bacterial inflammation of the tissues surrounding the nail. Redness, pus, and swelling are usually seen in the skin fold adjacent to the nail plate.
<i>Pseudomonas Aeruginosa</i> (SUE-duh-MOAN-us aye-ru-jin-oh-sa)	Common bacteria that can lead to a bacterial infection that appears as a green, yellow, or black discoloration on the nail bed.
Pyogenic Granuloma (py-oh-JEN-ik gran-yoo-LOH-muh)	Severe inflammation of the nail in which a lump of red tissue grows up from the nail bed to the nail plate.
Tinea Pedis (TIN-ee-uh PED-us)	Also known as <i>athlete's foot</i> ; red, itchy rash on the skin on the bottom of feet and/or between the toes, usually between the fourth and fifth toe.

Learn about Manicuring

After reading this section, you will be able to:

- LO3 Identify nail care equipment, implements, materials, and products.
- LO4 Recognize the five general shapes of nails.
- LO5 Understand table setup, hand massage, and manicure procedures.

The ancients regarded long, polished, and colored fingernails as a mark of distinction between aristocrats and common laborers. The word *manicure* is derived from the Latin *manus* (meaning “hand”) and *cura* (meaning “care”), which means the care of the hands and nails.

To perform professional manicures, it is important to develop competence when working with nail care tools. Nail care tools include equipment, implements, materials, and products.

figure A-3
Manicure table.



IDENTIFY MANICURE EQUIPMENT, IMPLEMENTS, MATERIALS, AND PRODUCTS

Equipment

Equipment used to perform nail services do not require replacement until they are no longer in good repair.

- *Manicure table with adjustable lamp:* Most standard manicuring tables include drawers for storage and an attached, adjustable lamp (Figure A-3). The lamp should have a 40- to 60-watt bulb. Heat from a higher-wattage bulb will interfere with manicuring and sculptured nail procedures. A lower-wattage bulb will not warm a client’s nails in a room that is highly air-conditioned. The warmth from the bulb will help maintain product consistency.
- *Client’s chair and technician’s chair or stool.*

- *Finger bowl:* A plastic, china, metal, or glass bowl is used for soaking the client's fingers in warm water and antibacterial soap (**Figure A-4**).
- *Disinfection container:* This receptacle must be large enough to hold the disinfectant solution in which to immerse implements for sanitizing purposes. A cover is provided with most containers to prevent contamination of the solution when it is not in use.
- *Client's arm cushion:* The cushion is usually 8 to 12-inches long and especially made for manicuring (a towel that is folded to cushion size can also be used). The cushion or folded towel should be covered with a clean towel before each appointment.
- *Gauze and cotton container:* This container holds clean, absorbent cotton, lint-free wipes, or gauze squares.
- *Supply tray:* The tray holds cosmetics such as polishers, polish removers, and creams. It should be sturdy and easy to clean.
- *Electric nail dryer:* A nail dryer is an optional item used to shorten the length of time necessary for drying the client's nails.
- *Metal trash can:* The trash can should have a foot pedal and lid and be lined with a plastic bag.
- *Approved solution for jar sanitizer:* Depending on state laws, *disinfected* metal implements may be placed in a small jar containing a disinfectant to maintain sanitary standards during the manicure. Always follow-up with thorough washing and disinfection of the implements and jar after each use. Check your state rules and regulations for further information.

Implements

Implements are tools that must be cleaned and disinfected or discarded after use with each client.

- *Wooden pusher:* Use a wooden pusher, also known as an *orangewood stick*, to loosen the cuticle around the base of a nail or clean under the free edge. Hold the stick similarly to a pencil (**Figure A-5**).
- *Metal pusher:* The metal pusher is used to push back excess cuticle growth. Hold the pusher in the same way as the wooden pusher. The spoon end is used to loosen and push back the cuticle. If the pusher has rough or sharp edges, use an abrasive board to dull them. This prevents digging into the nail plate.
- *Abrasive nail file:* An abrasive nail file is used to shape the free edge of natural or sculptured nails (**Figure A-6**). Abrasive files are available in different grits; the lower the grit number, the more aggressive its action. Most professional nail technicians use 7- or 8-inch nail files because some states do not allow shorter files to be used. When using a nail file, hold it with the thumb on one side of the handle and four fingers on the other side at an angle to the free edge. Emery boards are another type of abrasive file.
- *Nipper:* A nipper is used to trim away dead skin at the base of the nail (**Figure A-7**). To use the nipper, hold it in the palm of the hand with the blades facing the cuticle. Place the thumb on one handle and three fingers on the other handle, with the index finger on the screw to help guide the blade around the cuticle.

figure A-4
Finger bowl for manicures.



figure A-5
Wooden pusher.



figure A-6
Typical abrasive nail file.



CAUTION

Always prep abrasive boards by rubbing a clean board across the sharp edges before use.

figure A-7
Nail nipper.



figure A-8
Abrasive nail block buffer.



- *Tweezers:* Tweezers can be used to lift small particles from the nail.
- *Nail brush:* A nail brush is used to clean fingernails and remove bits of cuticle with warm, soapy water. Hold the nail brush with the bristles turned down and away from you. Place the thumb on the handle side of the brush facing you and the fingers on the other side.
- *Nail buffer:* Buffers used to be made with a chamois cover, but new disposable materials have replaced them. Two- or three-way buffers are used to add shine to the nail and to smooth out corrugations or wavy ridges (**Figure A-8**).
- *Fingernail clippers:* Fingernail clippers are used to shorten nails. For very long nails, clipping reduces filing time.
- *Plastic or metal spatula:* The spatula is used to remove nail cosmetics from their containers. Never use your fingers because you will transfer bacteria into the container and contaminate the product.

Materials

Materials are supplies that need to be replaced for each client.

- *Disposable towels or terry cloth towels:* A fresh, sanitized terry towel is used to cover the client's cushion before each manicure. Another fresh towel should be used to dry the client's hands after soaking in the finger bowl. Other terry or lint-free disposable towels are used to wipe spills that may occur around the finger bowl.
- *Cotton or cotton balls:* Cotton is used to remove polish, wrap the end of the wooden pusher, and apply nail cosmetics. Some nail technicians prefer to use small, fiber-free squares to remove polish because they do not leave cotton fibers on the nails that might interfere with polish application.
- *Gloves:* Use gloves when disinfecting implements and surfaces. Check with state board laws regarding required use during a manicure. Gloves are personal protective equipment (PPE) worn to protect the barber from exposure to microbes during services.
- *Plastic bags:* Tape or clip a bag to the side of the manicuring table to hold materials used during a service. Line all trash cans with plastic bags. Be sure to have a generous supply of bags so that they can be changed regularly during the day.

Products

This section describes some basic nail cosmetics that are often used during a manicure.

- *Antibacterial soap:* This soap is mixed with warm water and used in the finger bowl to clean.
- *Buffing powder or cream:* Buffing powder is used with a chamois buffer or three-way buffer to polish and add shine to the surface of the nail plate. The dry version may also be known as *pumice powder*.
- *Polish remover:* Polish remover is used to dissolve and remove nail polish; use non-acetone polish remover for clients who have artificial nails, since acetone will weaken or dissolve the tips, wrap glues, and sculptured nail compound.

- *Cuticle cream:* Cuticle cream is used to lubricate and soften dry cuticles and brittle nails.
- *Cuticle oil:* Cuticle oil keeps the cuticle soft and helps prevent hangnails or rough cuticles.
- *Cuticle solvent or cuticle remover:* Cuticle solvent makes cuticles easier to remove and minimizes nipper work.
- *Nail bleach:* Nail bleach is applied to the nail plate and under the free edge to remove yellow stains.
- *Nail whitener:* Nail whiteners are applied under the free edge of a nail to make the nail appear white.
- *Nail strengthener/hardener:* Nail strengthener is applied to the nail before the base coat to prevent splitting and peeling of the nail. Two types of nail strengtheners are protein hardeners and nylon fiber hardeners.
- *Base coat:* The base coat is colorless and is applied to the nail before the application of colored polish or a clear top coat unless a nail strengthener is being used.
- *Colored polish, liquid enamel, or lacquer:* Colored polish is used to add color and gloss to the nail and is usually applied in two coats.
- *Top coat or sealer:* The top coat, a colorless polish, is applied over colored polish to prevent chipping and to add a shine to the finished nail.
- *Liquid nail dry:* Liquid nail dry promotes rapid drying and is available in brush-on or spray form.
- *Hand cream and hand lotion:* Hand lotion and hand cream add a finishing touch to a manicure and help replace lost moisture to the skin.
- *Nail conditioner:* Nail conditioner contains moisturizers and helps prevent brittle nails and dry cuticles.

TYPES OF POLISH APPLICATIONS

When men choose to wear polish, it is usually a clear polish that is applied to the entire nail plate (full coverage). Women, on the other hand, may request one of the following coverage options instead of full coverage:

- *Free edge:* The free edge of the nail is unpolished. This helps prevent the polish from chipping.
- *Hairline tip:* The nail plate is *polished* and a $\frac{1}{16}$ inch is removed from the free edge. This prevents the polish from chipping.
- *Slim-line or free walls:* Leave a $\frac{1}{16}$ -inch margin on each side of the nail plate. This makes a wide nail appear narrower.
- *Half-moon or lunula:* The lunula at the base of the nail is left unpolished.

APPLYING POLISH

When applying polish, the base coat is applied first, followed by two coats of color and one or two applications of top coat. Roll the polish bottle in the palms of the hands to mix. Never shake polish. Shaking causes air bubbles to

form, which can be transferred to the nail plate during application and cause marks in the finish. Apply all coats of polish in the following manner:

1. Remove the brush from the bottle and wipe one side on the bottleneck so that a bead of polish remains on the end of the brush.
2. Start in the center of the nail, position the brush $\frac{1}{16}$ inch away from the cuticle, and brush toward the free edge.
3. Using the same technique, do the left side of nail, then the right side. There should be enough polish on the brush to complete three strokes without having to dip it back into the bottle; however, the amount will need to be adjusted according to the size of the nail. The more strokes used, the more lines or lumps will show on the client's nail. Small areas missed with the first color coat can be covered with the second coat.
4. Apply two coats of colored polish using the same technique used for the base coat. Complete the first color coat on both hands before starting the second coat. Polish on the cuticle should be removed with a cotton-tipped orangewood stick saturated in polish remover.
5. Apply one or two coats of top coat to prevent chipping and to give nails a glossy look. The use of an instant nail dry spray is optional, but it is effective in preventing smudging and dulling.

CHAIR-SIDE MANICURE

In some barbershops, the manicurist performs the manicure at the barber's workstation. This procedure is called a *chair-side* or *booth manicure* and requires the manicurist to either balance the supply tray on their lap or to have a small table at hand. If manicures are to be performed chair-side, the styling chair should have a small, recessed hole at the end of the armrest to hold the finger bowl. The manicurist must then move around the client, depending on which hand is being manicured.

When performing a chair-side manicure, always be considerate of the barber's position during the haircutting and styling process. Try to anticipate the turn of the barber chair or a change in the client's position in order to prevent client discomfort and any interference with the barber's procedures.

P A-1 Pre-service Procedure *pages 792–797*

P A-2 Post-service Procedure *pages 798–799*

CLIENT CONSULTATION

Before performing a service on a client, take time to talk with that client. During the client consultation, discuss issues of general health, the health of nails and skin, and the client's lifestyle and needs. You will use your knowledge of skin, nails, and each type of nail service to help the client select the most appropriate service. If the client has a nail or skin disorder that prevents you from performing a service, refer that client to a physician and offer to perform a service as soon as the disorder has been treated. A good client consultation can make a difference between being a professional and just doing nails.

Finish the consultation with a determination of the desired shape and polish color of the nails. Consider the shape of the hands, the length of the fingers,

the shape of the cuticles, and the type of work the client does. The five standard nail shapes are square, squoval, round, oval, and pointed (see [Table A-3](#)).

Most men prefer short nails. The square or round shapes can be accomplished on shorter nail lengths and usually look the most appropriate on male hands. Generally, tapering the nail ends to a point or oval shape should be reserved for use on female hands and nails.

Generally speaking, a woman's manicure is performed using the same steps as a man's manicure with the exception of Step 14 in Procedure A-4, which usually requires the application of a base coat, two coats of colored polish, and one or two coats of top coat.

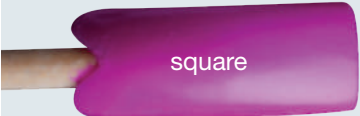
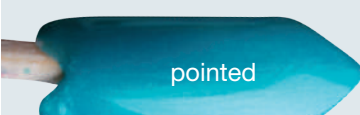
P A-3A Hand Massage Techniques *pages 800–801*

P A-3B Arm Massage Techniques *pages 801–802*

P A-4 Men's Manicure *pages 803–806*

table A-3

BASIC NAIL SHAPES

Shape	Definition
	The square nail is completely straight across the free edge with no rounding at the outside edges.
	The squoval nail has a square free edge that is rounded off at the corner edges. If the nail extends only slightly past the fingertip, this shape will be sturdy because there is no square edge to break off, and any pressure on the tip will be reflected directly back to the nail plate, its strongest area. Clients who work with their hands—nurses, computer technicians, landscapers, or office workers—will need shorter, squoval nails.
	The round nail should be slightly tapered and usually should extend just a bit past the fingertip.
	The oval nail is a conservative nail shape that is thought to be attractive on most women's hands. It is similar to a squoval nail with even more rounded corners. Professional clients who have their hands on display (e.g., businesspeople, teachers, or salespeople) may want longer oval nails.
	The pointed nail is suited to thin hands with long fingers and narrow nail beds. The nail is tapered and longer than usual to emphasize and enhance the slender appearance of the hand. Know, however, that this nail shape may be weaker, may break more easily, and is more difficult to maintain than other nail shapes. Rarely are natural nails successful with this nail shape, so they are usually enhancements. They are for fashion-conscious people who do not need the strongest, most durable shape of nail enhancements.

PRE-SERVICE PROCEDURE

MATERIALS, IMPLEMENTS, AND EQUIPMENT

- ☐ Buffing cream
- ☐ Cotton or pledgets
- ☐ Disinfecting container with disinfectant solution
- ☐ Disposable implements (abrasive file, wooden pusher, buffers)
- ☐ Finger bowl and brush
- ☐ Gloves
- ☐ Hand soap or sanitizer
- ☐ Linen or paper towels
- ☐ Manicure table and chairs
- ☐ Massage cream or lotion
- ☐ Metal implements (fingernail clippers, nippers, file, metal pusher)
- ☐ Metal trash receptacle with self-closing lid or small plastic bags and tape
- ☐ Polishes

PRE-SERVICE PROCEDURE

- 1 Apply gloves. Clean and disinfect reusable tools and implements and store in clean dry container until needed.
- 2 Clean and disinfect manicure table and drawer with an EPA-approved disinfectant.
- 3 Remove gloves and wash your hands.
- 4 Conduct client consultation.

PROCEDURES

A. CLEANING AND DISINFECTING



- 1 It is important to wear gloves while performing this pre-service procedure to prevent possible contamination of the implements by your hands and to protect your hands from the powerful chemicals in the disinfectant solution.



- 2 Rinse all implements with warm running water, and then thoroughly wash them with soap, a nail brush, and warm water. Brush grooved items, if necessary, and open hinged implements to scrub the area.



③ Rinse away all traces of soap with warm running water. The presence of soap in most disinfectants can cause the disinfectant to become inactive. Soap is most easily rinsed off in warm, but not hot, water. Dry implements thoroughly with a clean cloth towel or a disposable towel. Your implements are now properly cleaned and ready to be disinfected.



④ It is extremely important that your implements be cleaned before placing them in the disinfectant solution. Otherwise, your disinfectant may become contaminated. Before immersing the cleaned implements, bring any hinged implements to the open position. Immerse cleaned implements in a disinfection container holding an EPA-registered disinfectant for the required time (usually 10 minutes). Change the disinfectant solution daily or sooner if the disinfectant becomes visibly dirty during the course of the day. Avoid skin contact with all disinfectants by using tongs or by wearing disposable gloves.



⑤ Remove implements, avoiding skin contact, and rinse and dry tools thoroughly.



⑥ Store disinfected implements in a clean, dry container until needed.



- 7 Remove gloves and thoroughly wash your hands with liquid soap. Rinse and dry with a clean fabric or disposable towel.

B. BASIC TABLE SETUP



- 8 Following the directions on the product label, clean and then disinfect the manicure table and drawer with an EPA-approved disinfectant.



- 9 Wrap your client's arm cushion, if you are using one, with a clean cloth or disposable towel. Place the cushion in the middle of the table so that one end of the towel extends toward the client and the other end extends toward you.



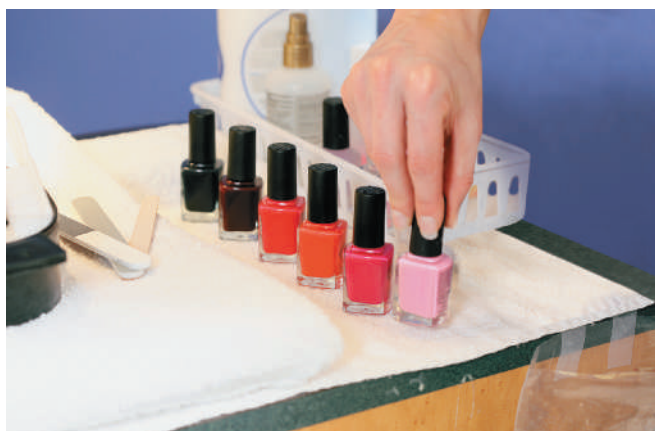
- 10 Place the abrasives and buffers of your choice on the table to your right (or to the left if you are left-handed). Many technicians wrap them neatly in a towel to ward off dust and potential contaminants.



- 11 Place the finger bowl filled with warm water and the manicure brush in the middle of the table. The finger bowl should not be moved from one side to the other of the manicure table. It should stay in place for the duration of your manicure.



12 Tape or clip a plastic bag that can be closed securely to the side of the table if a metal trash receptacle with a self-closing lid is not available. This is used for depositing used materials during your manicure. These bags must be sealed and thrown away after each client to prevent product vapors from escaping into the shop air.



13 Place polishes to the right if you are right-handed and to the left if you are left-handed.



14 The drawer can be used to store the following items in clean, covered containers for immediate use: extra cotton or cotton balls, abrasives, buffers, nail polish dryer, and other supplies. Never place used materials in your drawer. Only completely cleaned and disinfected implements in a clean, covered container (to protect them from dust and recontamination) are stored in the drawer; extra materials or professional products are also placed in a clean, covered container in the drawer. Your drawer should always be organized and clean.

C. GREET THE CLIENT



15 Greet your client with a smile, introduce yourself if you have never met, and shake hands. If the client is new, ask for the consultation card or sheet they filled out in the reception area.



16 Escort your client to the hand-washing area and demonstrate the hand-washing procedure on your own hands. Once you have completed the demonstration, hand your client a fresh nail brush to use and ask them to wash their hands.



17 Hand your client a fresh towel for drying their hands. Be sure that your towels are clean and are not worn. A dirty towel can cause a client to not come back or to report the shop to the state board.



18 Show your client to your manicure table and make sure they are comfortable before beginning the service.



19 Discuss the information on the consultation card and determine a course of action for the service.

POST-SERVICE PROCEDURE

PROCEDURES

A. ADVISE CLIENTS AND PROMOTE PRODUCTS



1 Proper home maintenance will ensure that the client's nails look beautiful until your client returns for another service (polish should last 7 to 10 days).



2 Depending on the service provided and the condition of your client's hands, there may be a number of retail products that you should recommend for the client to take home. This is the time to do so. Explain why they are important and how to use them.

B. SCHEDULE THE NEXT APPOINTMENT AND THANK THE CLIENT



3 Escort the client to the front desk to schedule the next appointment and pay for the service. Set up the date, time, and services. Write the information on your or the shop's appointment card and give it to the client.



4 Before the client leaves the shop, thank them for their business and mention that you will be looking forward to their next visit.



5 Record service information, products used, observations, and retail recommendations on the client service form or computer record.

C. PREPARE THE WORK AREA AND IMPLEMENTS FOR THE NEXT CLIENT



6 Remove your products and tools, dispose of all used materials, and then clean and disinfect your work area.



7 Follow the steps for disinfecting implements in the Pre-service Procedure.

HAND MASSAGE

PROCEDURES

A. HAND MASSAGE TECHNIQUES



1 *Relaxation movement:* This is a form of massage known as *joint movement*. Apply hand lotion or cream. Place the client's elbow on a cushion. With one hand, brace the client's arm. With the other hand, hold the client's wrist and bend it back and forth slowly 5 to 10 times or until you feel the client has relaxed.



2 *Joint movement on fingers:* Lower the client's arm, bracing the right hand so you can start the massage on the little finger. Hold the finger at the base of the nail and gently rotate to form circles. Work toward the thumb, three to five times on each finger.



3 *Circular movement in palm:* Use the effleurage manipulation. Place the client's elbow on the cushion and, with your thumbs in the client's palm, rotate in a circular movement in opposite directions.



4 *Circular movement on wrist:* Hold the client's hand between your hands, placing your thumbs on top and your fingers below the client's hand. Move the thumbs in a circular motion in opposite directions, from the client's wrist to his knuckles. Move up and down three to five times. At the last rotation, wring the client's wrist by bracing your hands around the wrist and gently twisting in opposite directions.



- 5 *Circular movement on back of the hand and fingers:* Rotate down the back of the client's hand using the thumbs. Rotate down the little finger and the client's thumb, and gently squeeze off at the tips of the client's fingers. Go back and rotate down the ring finger and index finger, gently squeezing off. Now do the middle finger and squeeze off at the tip. This tapering to the fingertips helps blood flow.

B. ARM MASSAGE TECHNIQUES



- 6 Warm cream or lotion in your hands, apply to the client's arm and work it in. Work from the client's wrist toward the elbow, except on the last movement, when work should be from the elbow to the wrist. Finally, squeeze off at the fingertips, as at the end of a hand massage. Apply more cream if necessary.



- 7 *Effleurage on arms:* Put the client's arm on the table, bracing the arm with your hands. Hold the client's hand palm up in your hand. Your fingers should be under the client's hand, your thumbs side by side in the client's palm. Rotate your thumbs in opposite directions, starting at the client's wrist and working toward the elbow. When you reach the elbow, slide your hand down the client's arm to the wrist and rotate back up to the elbow three to five times. Turn the arm over and repeat three to five times on the top side of arm.



8 *Friction massage movement (wringing movement):* A friction massage involves deep rubbing of the muscles. Bend the client's elbow so the arm is horizontal in front of you, with the back of the hand facing up. Place your hands around the arm with your fingers facing in the same direction as the arm, and gently twist in opposite directions as you would wring out a washcloth, from wrist to elbow. Repeat up and down the forearm three to five times.



9 *Kneading movement on the arm:* Place your thumbs on the top side of the client's arm so they are horizontal. Move them in opposite directions, from wrist to elbow and back down to wrist. This squeezing motion moves the flesh over the bone and stimulates the arm tissue. Do this three to five times.



10 *Rotation of the elbow, friction massage movement:* Brace the client's arm with your left hand and apply cream to the elbow. Cup the elbow with your right hand and rotate your hand over the client's elbow. Repeat three to five times. To finish the elbow massage, move your left arm to the top of the client's forearm. Gently slide both hands down the forearm from the elbow to the fingertips as if climbing down a rope. Repeat three to five times.

MEN'S MANICURE

Manicure procedures are basically the same for men and women. Table setup is the same, except that colored polish is not typically used on men. Some men like a clear liquid polish, while others prefer a dry or buffed finish. Using hand cream or lotion is optional and depends on the client's preference. The following is one method of performing a men's manicure.

MATERIALS, IMPLEMENTS, AND EQUIPMENT

- ☐ Buffing cream
- ☐ Clear polish
- ☐ Cotton or pledgets
- ☐ Disinfection container with disinfectant solution
- ☐ Disposable implements (abrasive file, wooden pusher, buffers)
- ☐ Finger bowl and brush
- ☐ Gloves (if required by regulations)
- ☐ Hand soap or sanitizer
- ☐ Linen or paper towels
- ☐ Manicure table and chairs
- ☐ Massage cream or lotion
- ☐ Metal implements (fingernail clippers, nippers, file, metal pusher)
- ☐ Metal trash receptacle or plastic bags and tape

PREPARATION

- 1 Perform pre-service and table setup procedures.
- 2 Greet the client.
- 3 Wash your hands. Have the client wash his hands or apply a hand sanitizer. Thoroughly dry hands and nails with a sanitized towel. Apply gloves if required.
- 4 Perform consultation and discuss service options with the client.
- 5 Begin working with client's left hand so you can work from right to left.

PROCEDURE

- 1 Remove clear polish if applicable. Begin with the little finger of the left hand, using cotton saturated with polish remover. Repeat on the right hand.



- 2 Shape the nails. Most men keep their nails fairly short; if the nails are long, shorten them with fingernail clippers before filing. File from the corners to center in one direction; do not saw back and forth.



3 Soften the cuticles. After filing the nails of the left hand, soak them in a soap bath while filing the second hand.



4 Clean the nails. Use a nail brush to clean fingertips and nails and to remove surface debris from nails. Remove the left hand from the soap bath while placing the right hand into the bath, and brush the fingers with downward strokes, starting at the first knuckle and brushing in one direction toward the free edge.



5 Dry the hand. Use the end of a clean towel, making sure to dry between the fingers. As you dry, gently push back the eponychium with the towel.



6 Apply cuticle remover. Use a cotton-tipped wooden pusher to apply cuticle remover to each nail on the left hand. Put client's left hand back into the bath while repeating cleansing steps on the right hand. Apply cuticle remover to right hand, remove left hand from bath, and replace with right hand.



7 Loosen the cuticles. Use the spoon end of the pusher or a wooden pusher to gently push back and lift the cuticle off the nails of the left hand. Repeat on right hand.



8 Use nippers, if allowed in your state, to remove any loosely hanging tags of dead skin only; do not cut living skin.



9 Clean under the free edge with a cotton-tipped wooden pusher while holding the left hand over the soap bath, and brush a last time to remove bits of cuticle and nail debris that remain on the nail. Rest the left hand on a clean towel and repeat this procedure on the right hand.



10 Apply buffing powder or cream and buff.

11 Apply cuticle oil. Use a cotton-tipped wooden pusher, a cotton swab, or an eyedropper to apply and massage oil into the nail plate and surrounding skin using a circular motion.

12 Bevel nails if necessary.



13 Apply hand lotion and massage the hands and wrists; wipe excess lotion off nails.



14 Polish the nails with a clear matte polish if desired or re-buff gently.



15 Complete post-service procedure.

POST-SERVICE PROCEDURE

- 1** Schedule another appointment with the client to maintain the manicure or to perform another service; complete client's record card.
- 2** Clean up work area. Disinfect table, tools, and implements. Take the time to restore the basic table setup.
- 3** Place all used materials in the plastic bag at the end of the table and discard into a closed trash can.
- 4** Wash your hands.

REVIEW QUESTIONS

- 1 Briefly describe each of the following parts of the nail: nail plate, nail bed, matrix, cuticle, eponychium, perionychium, hyponychium, specialized ligaments, and nail folds.
- 2 What is the average growth rate of a nail?
- 3 Name two common nail disorders that may be serviced by a manicurist or nail technician?
- 4 List the supplies needed to perform a *men's* manicure.
- 5 What are the five basic nail shapes?
- 6 Identify nail shapes that look appropriate on a man's hand.
- 7 Name five hand massage techniques and four arm massage techniques.
- 8 Briefly list the steps of a basic men's manicure procedure.
- 9 Explain the manner in which a chair-side manicure is performed.

CHAPTER GLOSSARY

bed epithelium (BED ep-ih-THÉE-lee-um)	p. 782	thin layer of tissue between the nail plate and the nail bed
cuticle (KYOO-tih-kul)	p. 783	dead, colorless tissue attached to the nail plate around the base of the nail
discolored (dis-KUL-urd)	p. 784	a condition in which the nails turn a variety of colors such as yellow, blue, blue-gray, green, red, or purple; can be caused by poor blood circulation, a heart condition, topical or oral medications, or a systemic disorder
eponychium (ep-oh-NIK-ee-um)	p. 783	living skin at the base of the nail plate and covering the matrix area
free edge (FREE EJ)	p. 782	part of the nail plate that extends over the tip of the finger
hangnails (HANG-NAYLZ)	p. 784	condition in which the cuticle splits around the nail
hyponychium (hy-poh-NIK-ee-um)	p. 783	slightly thickened layer of skin between the fingertip and free edge of the nail plate
lunula (LOO-nuh-luh)	p. 783	half-moon shape at the base of the nail
matrix (MAY-triks)	p. 783	area where the nail is formed; produces cells that create the nail plate
nail (NAYL)	p. 782	an appendage of the skin; horny protective plate at the end of the finger or toe
nail bed (NAYL BED)	p. 782	portion of the skin on which the nail plate rests
nail folds (NAYL FOHLDZ)	p. 783	folds of normal skin around the nail plate
nail grooves (NAYL GROOVZ)	p. 783	slits or furrows on the sides of the nail
nail plate (NAYL PLAYT)	p. 782	horny plate resting on and attached to the nail bed
onyx (AHN-iks)	p. 782	technical term for nail
perionychium (payr-ee-uh-NIK-ee-um)	p. 783	the living skin bordering the root and sides of a nail
ridges (RIJ-ez)	p. 782	depressions running vertically down the length of the nail
specialized ligaments	p. 783	bands of fibrous tissue that attach the nail bed and matrix to the underlying bone that are located at the base of the matrix and around the edges of the nail bed

GLOSSARY / INDEX

A

ABCDE Cancer Checklist, 240

Abductors, muscles that draw a body part, such as a finger, arm, or toe, away from the midline of the body or of an extremity, 156–157

Absorption, 228, 229

AC. *See* Alternating current (AC)

Accelerated hydrogen peroxide (AHP), 74

Accessory nerve, 160, 308

Achievement-oriented resume, 719, 720

Acid-balanced shampoos, 193

Acid-balanced waves, permanent waves that have a 7.0 or neutral pH; do not require hair-dryer heat, 597

Acids, solutions that have a pH below 7.0, 184, 189–190

Acne, skin disorder characterized by chronic inflammation of the sebaceous glands from retained secretions, 234, 325

Acne rosacea, 235

Acne treatment, 235

Acquired immune deficiency syndrome, abbreviated AIDS; a disease that breaks down the body's immune system; AIDS is caused by the human immunodeficiency virus (HIV), 70

Acquired immunity, immunity that the body develops after overcoming a disease, through inoculation (such as flu vaccinations) or through exposure to natural allergens (such as pollen, cat dander, and ragweed), 71

Acronyms, 32

Activator, an oxidizer, consisting of powdered persulfate salts, that is added to haircolors, lighteners, or hydrogen peroxide to increase the chemical action of the product, 656

Active stage, 66

Adductors, muscles that draw a body part, such as a finger, arm, or toe, inward toward the median axis of the body or of an extremity, 156–157

Adipose tissue, a technical term for fat; gives smoothness and contour to the body, 147, 225

Adjustable-blade clippers, 115

Adult clients, safe work practices and safety precautions, 83–84

Advertising, 776–778

Aerosol cleaner, 74, 91

Afferent nerves, 158

African cultures, 7

AIDS. *See* Acquired immune deficiency syndrome (AIDS)

Albinism, congenital leukoderma, or absence of melanin pigment in the body, 238, 260

Alcohol, 199

Alexander the Great, 10

Alkaline or cold waves, perms that process at room temperature without heat with a pH range between 9.0 and 9.6, 596

Alkalis, solutions that have a pH above 7.0, 184, 189–190

Alkanolamines, 200

Allergy, reaction due to extreme sensitivity to certain foods, chemicals, or other normally harmless substances, 68, 73

All-purpose combs, 106

Alopecia, abnormal hair loss, 258, 284, 299

Alopecia areata, the sudden falling out of hair in patches or spots—autoimmune disorder, 258

Alopecia prematura, hair loss that occurs before middle age, 258

Alopecia senilis, hair loss occurring in old age, 259

Alopecia syphilitica, hair loss as a result of syphilis, 259

Alopecia totalis, total loss of scalp hair, 259

Alopecia universalis, complete loss of body hair, 259

Alternating current (AC), rapid and interrupted current, flowing first in one direction and then the opposite direction, 207–208

Alternative hair replacement methods, 497–499

Alum, 200

AMBBA. *See* Associated Master Barbers and Beauticians of America (AMBBA)

Americans with Disabilities Act (ADA), 772

Amino acids, the building blocks or units of structure in protein, 250

Ammonia, 200

Ammonia-free waves, 597–598

Ammonium thioglycolate (ATG), main active ingredient or reducing agent in alkaline waves, 183, 200, 587, 596, 599, 607

Ampere (or amp) (A), standard unit for measuring the strength of an electric current or the rate of flow of charge in a conductor; also called an *amp*, 208

Amphoterics, 192

Anagen phase, growth phase in the hair cycle, 256

Anaphoresis, process of forcing substances into tissues using galvanic current from the negative toward the positive pole, 212, 317

Anatomical features and haircutting

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facial shapes, 387–389

head shapes, 389

neck lengths, 390–391

profiles, 390

sideburn lengths and designs, 391–392

Anatomy, the science of the structure of organisms and of their parts, 144

body systems and associated organs, 147–167

cells, 145–146

circulatory system, 161–165

endocrine system, 166–167

integumentary system, 165–166

lymphatic/immune system, 165

muscular system, 151–157

nervous system, 157–160

reproductive system, 167

of skin, 222–229

tissues, types of, 147

Androgenic alopecia, hair loss that occurs as a result of genetics, age, and hormonal changes; male pattern baldness, 258

Angle, the space between two lines or surfaces that intersect at a given point; in haircutting, the hair is held away from the head to create an angle of elevation, 397

Angular artery, artery that supplies blood to the sides of the nose, 164

Angular profiles, 390

Anhidrosis, a deficiency in perspiration or the inability to sweat, often a result of damage to autonomic nerves, 236

Aniline derivative, uncolored dye precursors that combine with hydrogen peroxide to form larger, permanent color molecules in the cortex; the ingredient that requires a patch test, 650

Aniline derivative tints, 674

Anion, 188

Anionics, 191–192

Anode, positive electrode, 211–212

Anterior auricular artery, artery that supplies blood to the front part of the ear, 164

Antimicrobial and antibacterial soaps, 79

Antiseptics, chemical germicide formulated for use on skin; registered and regulated by the Food and Drug Administration, 79

Aorta, largest artery in the body, 161

Arching, method used to cut around the ears and down the sides of the neck

with outliner, 425–426

with razor, 426–427

with shears, 424–425

Arching with clipper or trimmer, 407–408

Arm bones, 151

Arm muscles, 155–156

Arrector pili, involuntary muscle fiber attached to the base of the hair follicle, 249

Arteries, thick-walled, muscular, flexible tubes that carry oxygenated blood away from the heart to the arterioles, 161, 305

Arterioles, small arteries that deliver blood to capillaries, 161

Ash or drab colors, 646

Associated Master Barbers and Beauticians of America (AMBBA), 15

Associated Master Barbers of America, 15

Asteatosis, a condition of dry, scaly skin, characterized by the absolute or partial deficiency of sebum, 235

Astringents, tonic lotions with an alcohol content of up to 35 percent; used to remove oil accumulation on oily and acne-prone skin, 197, 322

Asymptomatic, showing no symptoms or signs of infection, 80

Atoms, are the basic building blocks of all matter and are the smallest particle of an element that has the chemical identity of the chemical, 180

Attitude, 36

Auricularis anterior, muscle in front of the ear that draws the ear forward, 153

Auricularis posterior, muscle behind the ear that draws the ear backward, 153

Auricularis superior, muscle above the ear that draws the ear upward, 153

Auriculotemporal nerve, nerve that affects the external ear and skin above the temple, up to the top of the skull, 159

Autoclave, 72

Average porosity, 643

Axon, extensions of a neuron through which impulses are sent away from the cell body to other neurons, glands, or muscles, 158

Axon terminal, 158

B

Bacilli, short rod-shaped bacteria; they are the most common bacteria and produce diseases such as tetanus (lockjaw), typhoid fever, tuberculosis, and diphtheria, 65

Back and forth movement, 282

Back bones, 150

Back injuries, 45

Backhand, razor position and stroke used in 4 of the 14 basic shaving areas: nos. 2, 6, 7, and 9; optional position for area 12, 342–343, 345, 356

Bacteria, single-celled microorganisms that have both plant and animal characteristics; some bacteria are harmful; some are harmless, 63–67

Bacterial infections, 66–67, 264–265

Bacterial spores, bacteria capable of producing a protective coating that allows them to withstand very harsh environments and to shed the coating when conditions become more favorable to them, 63

Bactericidal, capable of destroying bacteria, 63

Balance, the equal or appropriate proportions that create symmetry and harmony in a design, 396

Balancing shampoos, are designed for oily hair and scalps. These shampoos wash away excess oiliness while keeping the hair from drying out, 193, 194

Barba, Latin for beard, 8

Barber chairs, 126–127

Barber code of ethics, 15

Barber law, 712–713

Barber pole, symbol of Barbering that evolved from the technical procedures of bloodletting performed by barber-surgeons, 12–13

Barbering

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employee evaluation, 742

employment classifications, 738–739, 741

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personnel, 770–772

reception area, 772

receptionist, 772–773

selling in, 778

success, elements of, 768–775

teamwork, 736–737

use of telephone in, 774–775

Barbershop operation, the ongoing, recurring processes or activities involved in the running of a barbershop for the purpose of producing income and value, 760

Barbershop policies, the rules or regulations adopted by a barbershop to ensure that all clients and associates are being treated fairly and consistently, 760

Barbershop visit, 723–724

Barber-surgeons, early practitioners who cut hair, shaved, and performed bloodletting and dentistry, 11–13

Basal cell carcinoma, most common and least severe type of skin cancer, 239

Basal cell layer, 224

Base, the stationary foundation of the curl on which the barrel (or roller) is placed, 541, 542

Base color, the predominant tone of an existing color, 647

Base control, the position of the perm rod in relation to its base section, 593

Base cream, protective cream used on the scalp during hair relaxing, 606–608

Base direction, angle at which the perm rod is positioned on the head; also, the directional pattern in which the hair is wrapped, 593–594

Base relaxers, relaxers that require the use of a base or protective cream, 606

Base sections, subsections of panels into which the hair is divided for perm wrappings; one rod is normally placed on each base section, 592

Bases, also known as *alkalis*, 184

Bases of hair replacements, 500–501
 Basic facial, 324
 Basic haircuts, women's, 532–535
 Basic taper cut, 439–443
 Beard design, 351, 375–380
 Beards, coloring, 674, 699–701
Bed epithelium, thin layer of tissue between the nail plate and the nail bed, 782
Belly, middle part of a muscle, 151, 152
 Benzoyl peroxide, 235
 Beveled edge, 109–110
Bicep, muscle that produces the contour of the front and inner side of the upper arm; lifts the forearm and flexes the elbow, 155
Binary fission, the division of bacterial cells into two new cells called *daughter cells*, 66
Biofilms, colonies of microorganisms that adhere to environmental surfaces, as well as the human body, 69
 Birthday cards, 750
 Birthmark, 238
Blackhead, an open comedone; consists of an accumulation of excess oil (sebum) that has been oxidized to a dark color, 228
 Blade edges. *See* Shears
 Blade wash, 73, 90
Blades, the cutting parts of the clippers, usually manufactured from high-quality carbon steel and available in a variety of styles and sizes, 117
 Bleach, 75–76
 Blending areas, 393–394
 Blending shears, 111–112
Blood, nutritive fluid circulating through the circulatory system that supplies oxygen and nutrients to cells and tissues and removes carbon dioxide and waste from them, 162–163
 Blood circulation, 249
Blood vessels, tubelike structures that include the arteries, arterioles, capillaries, venules, and veins; function is to transport blood to and from the heart and then to various tissues of the body, 161
Bloodborne pathogens, disease-causing microorganisms that are carried in the body by blood or body fluids, such as hepatitis and HIV, 69–70
 Blowdryers, 129–130
 Blow-waving, 478
Blowdry styling, technique of drying and styling damp hair in one process, 418–419, 541, 568–570
 Blowdrying techniques, 477–478
Blunt cut, haircut in which all the hair comes to one point at 0 elevation to form a weight line, 532–533, 544–547
 Body movement, 45–47

Body positioning for shaving, 347
Body systems, groups of body organs acting together to perform one or more functions, 147–167
Bookend wrap, perm wrap in which an end paper is folded in half over the hair ends, 589–592
 Booking appointments, 773, 774–775
Booth rental, also known as *chair rental*; a form of self-employment, business ownership, and tax designation, distinguished by renting a booth or station in a barbershop, 739, 741, 767–768
 Braids, 420, 479–482
Brain, largest and most complex nerve tissue; part of the central nervous system contained within the cranium, 157
 Brand identity, 757
Bromhidrosis, foul-smelling perspiration, 236
 Brush cuts, 412
Brush machine, an electrical appliance with interchangeable brushes that is used to mechanically cleanse, stimulate, and exfoliate the skin surface, 314
 Brushes, 107–108
Buccal nerve, nerve that affects the muscles of the mouth, 160
Buccinator muscle, thin, flat muscle of the cheek between the upper and lower jaws, 154
 Budget worksheet, 744
Bulla, large blister containing a watery fluid, 230
 Bumper, 110
 Business card referrals, 751
 Business feasibility, 758
 Business name, 758
Business plan, a written description of your business as you see it today and as you foresee it in the next 5 years (detailed by year), 759, 762–763
Business regulations and laws, any and all local, state, and federal regulations and laws that you must comply with when you decide to open your barbershop or rent a booth, 759–760
 Business timeline, 758

C

Calcium hydroxide relaxers, 606
 Cancer, skin, 237, 239–240
 Caninus muscle, 154
Canities, technical term for gray hair, 260–261
Canvas strop, usually one side of a combination strop made of linen or silk, woven into a fine or coarse texture that removes metal burrs or imbrications left after honing, 125

Cap frost, lightening technique that involves pulling strands of hair through a perforated cap with a plastic or metal hook, 663, 672
Cap technique, coloring or lightening technique that involves pulling strands of hair through a perforated cap with a plastic or metal hook, 663
 Capes, 77–78, 127
Capillaries, tiny, thin-walled blood vessels that connect the smaller arteries to venules and bring nutrients to the cells and carry away waste materials, 161
Capital, money needed to invest in a business, 761
 Carbon atom, 147, 179
Carbuncle, the result of an acute, deep-seated bacterial infection in the subcutaneous tissue, 265
 Cardiac muscle, 151
 Cardiovascular system, 161–165
 Carpal tunnel syndrome, 45, 150
Carpus, the bones of the wrist, 151
 Carving, 539
Cast shears, shears that have been made by heating steel to a liquid form for pouring into a mold, 108
Catagen phase, transition phase of the hair growth cycle that signals the end of the growth phase, 256
Cataphoresis, process of forcing acidic substances into tissues using galvanic current from the positive toward the negative pole, 212, 317
Cathode, negative electrode, 211–212
 Cation, 188
 Cationics, 191–192
Cell membrane, part of the cell that encloses the protoplasm; permits soluble substances to enter and leave the cell, 146
 Cell reproduction and division, 146
Cells, basic units of all living things, 145–146
 Centers for Disease Control and Prevention (CDC), 72
Central nervous system (CNS), division of nervous system consisting of the brain, spinal cord, spinal nerves, and cranial nerves, 157
Centrioles, two small structures near the nucleus that move to each side during the mitosis process to help divide the cell, 146
Cervical cutaneous nerve, nerve located at the side of the neck; affects the front and sides of the neck to the breastbone, 160
 Cervical nerves, 160
Cervical vertebrae, seven bones that form the top part of the spinal column in the neck region, 150
 Chair cloths, 278
 Chair rental, 739

Changeable-blade straight razor, a type of straight razor that uses changeable, disposable blades, 119–120

Cheekbones, 150

Chemical blowout, partially straightens the hair with the intent that it will be picked out and cut, 610–611

Chemical change, change in the chemical composition of a substance by which new substances are formed, 181–182

Chemical compounds, combinations of two or more atoms of different elements united chemically with fixed chemical composition, definite proportions, and distinct properties, 184

Chemical curl reformations, 609

Chemical exfoliants, 322

Chemical hair relaxing, the process of rearranging the basic structure of extremely curly hair into a straightened form, 605–608

Chemical waving, *see also* permanent waving, 578–579

acid-balanced waves, 597

alkaline perms, 596

ammonia-free waves, 597–598

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true acid waves, 596–597

types of, 596–599

water vs. lotion wrapping, 594–595

wave wrap, 588–595

wrapping patterns, 595

wrapping/rodding techniques, 594

Chemical properties, characteristics that can only be determined with a chemical reaction, 181

Chemical services, draping methods for, 278

Chemical texture services, hair services that cause a chemical change that permanently alters the natural wave pattern of the hair, 578–579

chemical curl reformations, 609

chemical hair relaxers, 605–608

chemical waves, 588–605

chemistry of, 585–587

client consultation and analysis, 579–585

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physical and chemical actions of, 587

scalp and hair analysis, 580–581

test curl for permanent wave, 612–613

texturizers and chemical blowouts, 610–611

thio relaxer, 619–625

Chemicals, safe work practices and safety precautions, 84

Chemistry, the science that deals with the composition, structure, and properties of matter, 179

basic, 178–197

chemical compounds, 184

conditioners, 194–196

cosmetic preparations, 190–200

hair, chemical composition of, 250–255

matter, 179–180

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properties of water and pH, 187–190

pure substances and mixtures, 183–184

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states of matter, 181

styling aids, 196–197

United States Pharmacopeia (USP), 196–197

Chest bones, 150

Chewing muscles, 153

Children, safe work practices and safety precautions, 83

Chloasma, non-elevated spots due to increased pigmentation in the skin, 238

Cicatrix, also known as scar, slightly raised or depressed mark on the skin formed after an injury or a lesion of the skin has healed, 233

Cilia, hairlike extensions that protrude from cells and help sweep away fluids and particles, 65

Circle, also known as the curl; part of a curl that forms a complete circle, 541

Circuit breaker, switch that automatically interrupts or shuts off an electric circuit at the first sign of overload, 209

Circulatory system, system that controls the steady circulation of blood through the body by means of the heart and blood vessels, 148

blood, 162–163

blood vessels, 161

head, face, and neck arteries, 163–164

head, face, and neck veins, 164–165

heart, 161

Clarification, 49

Clarifying shampoos, shampoo with an active chelating agent that binds to and helps remove metals from the hair; also provides thorough cleansing and cuts through product buildup, 193

Classic Caesar, 414–415

Classic pompadour, 414

Clay masks, 323

Cleaning, a mechanical process (scrubbing) using soap and water or detergent and water to remove all visible dirt, debris, and many disease-causing germs; cleaning also removes invisible debris that interferes with disinfection; cleaning is what barbers are required to do before disinfecting, 56, 63, 86–88, 89–92

Cleansing creams, 197

Cleansing lotions, 197

Clear layer, 223

Client base, customers who are loyal to a particular barber, 745, 750–752

Client consultation, 385–387, 495, 579–585, 665

Clipper cutting techniques, 405–408

Clipper-over-comb, cutting over a comb with the clippers, 405, 407

Clippers, electric haircutting tools with a single adjustable-blade or detachable-blade system, 114

blades and guards, 117

cleaning and disinfecting, 89–92

history of, 20

holding, 117–118

motors, 116

parts of, 115–116

Close shaving, the procedure of shaving facial hair against the grain during the second-time-over shave, 349

Closed comedo, also known as a whitehead; a small bump just under the skin surface with white or cream colored sebum, 234

Clothing, 82–83, 726

CNS. *See* Central nervous system (CNS)

Coarse hair, 266, 643–644

Coating dyes, 675–676

Cocci, round-shaped bacteria that appear singly (alone) or in groups; the three types of cocci are staphylococci, streptococci, and diplococci, 65

COHNS elements, 250–251

Cold receptor, 226

Collagen, fibrous protein that gives the skin form and strength, 226

Color, 397

- Color fillers, equalize porosity and deposit color in one application to provide a uniform contributing pigment on pre-lightened hair,** 660
- Color oil lighteners, 658
- Color restorers, 654
- Color rinses, 650
- Color theory
base color, 647
complementary colors, 645–646
hue and tone, 646
laws of color, 644
level, 646–647
natural level and tone, 647–648
primary colors, 645
saturation, 647
secondary colors, 645
tertiary colors, 645
- Color wheel, 645–646
- Color-enhancing shampoos, created by combining surfactant bases with basic dyes,** 193, 650
- Combination bones, 123
- Combination skin, 321
- Combination strop, 125
- Combs, 105–106, 113
- Comedone, a mass of hardened sebum and skin cells in a hair follicle that may be open (blackhead) or closed (whitehead),** 234
- Comedone extractor, an implement used to extract blackheads,** 132
- Commission, a percentage of the revenue that the barbershop takes in from services performed by an employee, usually offered to that employee once the individual has built up a loyal clientele,** 740
- Common carotid arteries, arteries that supply blood to the head, face, and neck,** 163, 309
- Communicable, able to be communicated; transferable by contact from one person to another as in a communicable disease,** 67
- Communication, 36
- Communication skills, 47–50, 716
- Company of Barber-Surgeons, 11, 12
- Complaints, handling, 775
- Complementary colors, a primary and secondary color positioned opposite each other on the color wheel,** 645–646
- Complete electric circuit, the path of an electric current from the generating source through the conductor and back to its original source,** 207
- Compound dyes, 654
- Compound molecules, two or more atoms of different elements united chemically with fixed chemical composition, definite proportions, and distinct properties,** 180
- Concave grind, 121
- Concave profiles, 390
- Conditioner fillers, 660
- Conditioners, chemical agents used to deposit protein or moisturizers in the hair,** 190–192, 194–196, 277
- Conditioning packs, 195
- Conditioning shampoos, also known as moisturizing shampoos, are mild cream shampoos that contain moisturizing agents (humectants) designed either to “lock in” the moisturizing properties of the product or to draw moisture into the hair,** 193
- Conductor, any substance, medium, or material that conducts electricity,** 207
- Conductor cords, 92
- Conflict, 737
- Connective tissue, fibrous tissue that binds together, protects, and supports the various parts of the body; examples include bone, cartilage, ligaments, tendons, blood, lymph, and adipose tissue,** 147
- Construction of hair replacement, 500–501
- Consumption supplies, supplies used in the daily business operation,** 766
- Contagious disease, also known as communicable disease; disease that is spread from one person to another person; some of the more contagious diseases are the common cold, ringworm, conjunctivitis (pinkeye), viral infections, and natural nail or toe and foot infections,** 67, 68
- Containers, disinfecting, 77
- Contamination, the presence, or the reasonably anticipated presence, of blood or other potentially infectious materials on an item’s surface or visible debris or residues such as dust, hair, and skin,** 68
- Continuing education, 738
- Contraindication, any product, procedure, or treatment that should be avoided because it may cause undesirable side effects or be harmful to the individual,** 319
- Contributing color, 659
- Contributing pigment, pigment that lies under the natural hair color that is exposed when the natural color is lightened,** 644
- Conventional (Marcel) iron, 130, 131
- Conventional straight razor, a razor made of a hardened steel blade that requires honing and stropping to produce a cutting edge,** 119–120, 121–122, 123, 133, 135
- Converter, an apparatus that changes DC to AC,** 207
- Convex edge, 109–110, 111
- Convex profiles, 390
- Cool colors, 646
- Cordless clippers and outliners, 116
- Corium, 224
- Corporation, an ownership structure controlled by one or more stockholders,** 761
- Corrective hair treatment, 284
- Corrective haircolor, 672
- Corrective treatments, 320
- Corrugator muscle, facial muscle that draws eyebrows down and wrinkles the forehead vertically,** 154
- Cortex, middle layer of the hair shaft,** 250, 252–253
- Cosmetic preparations, 197–199
- Cover letter, 725
- Cowlick, tuft of hair that stands straight up,** 257
- Crane handle design, 111
- Cranium, an oval, bony case that protects the brain,** 149
- Crayons, 650–651, 674
- Cream lighteners, 658, 670
- Cream masks, 323
- Cream peroxides, 655
- Creativity, 27
- Crest, the widest area of the head, also known as the parietal ridge, temporal region, hatband, or horseshoe,** 393
- Crew cut, 412, 453–454
- Croquignole perm wrap, rodding from the hair ends to the scalp,** 594
- Cross bonds, 251
- Cross-checking, the process of parting off subsections opposite from the elevation or direction at which they were cut to check the precision of cutting lines or blending,** 398
- Crust, dead cells that have accumulated over a wound while healing,** 232
- Cultivated locks, 420–421
- Curl, the hair that is wrapped around the barrel of an iron or a roller,** 254, 257, 268, 536–537, 541–542, 583
- Curl reduction, 603–604
- Curl reformation, a soft-curl permanent; combination of a thio relaxer and thio permanent, whereby the hair is wrapped on wave rods; used to make existing curl larger and looser,** 609, 632–637
- Curling irons, 130
- Curly hair, 254, 536–537
- Curved lines, lines that move in a semicircular or circular direction,** 395, 396
- Custom hair solutions, 501
- Cuticle, cuticle, outermost layer of the hair shaft, 249–250; dead, colorless tissue attached to the nail plate around the base of the nail, 782–783, 787–791, 804–805**

Cutis, 224

Cutting above the fingers, cutting of the hair is performed on the outside of the fingers; used with horizontal or vertical 90-degree projections of hair, 403

Cutting below the fingers, cutting of the hair is below or performed on the inside of the fingers; used in 0- and 45-degree elevation cutting, 403

Cutting blade, 115

Cutting edges, 109

Cutting line, the position of the fingers when cutting a section of hair, 400

Cutting palm-to-palm, cutting of the hair is performed with the palms of both hands facing each other; an alternative method used in layering vertical or horizontal sections of hair, 403

Cutting stroke, the correct angle of cutting the beard with a straight razor, 342

Cyst, a closed, abnormally developed sac containing fluid or morbid matter, above or below the skin, 230

Cysteine, an amino acid with a sulfur atom that joins together two peptide chains, 252

Cytoplasm, is the watery fluid that surrounds the nucleus of the cell and is needed for growth, reproduction and self-repair, 145

D

Damaged hair, reconditioning, 673–674

Dandruff, 261–262, 284, 299

Daughter cells, 66, 146

DC. *See* Direct current (DC)

Debt repayment, 743

Decolorization, 657

Decontamination, 68

Deductive reasoning, the process of reaching logical conclusions by employing logical reasoning, 710–711

Defaulting, 743

Deltoid, a large, triangular muscle covering the shoulder joint that allows the arm to extend outward and to the side of the body, 155

Demipermanent haircolor, deposit-only haircolor product similar to semi-permanent but longer lasting, 649, 651–652, 668, 684–685

Demographics, information about a specific population including data on race, age, income, and educational attainment, 759

Dendrites, treelike branchings of nerve fibers extending from the nerve cell that carry impulses toward the cell and receive impulses from other neurons, 158

Density, 267, 535

Dental plaque, 69

Depilatories, 197–198

Depressor labii inferioris muscle, muscle surrounding the lower lip, 154

Derma, technical name for skin; also another name for the dermis, 224

Dermal papilla, small, cone-shaped elevation located at the base of the hair follicle that fits into the hair bulb, 249

Dermatitis, an inflammatory condition of the skin, 236

Dermatitis venenata, also known as *irritant contact dermatitis*; an eruptive skin condition due to contact with irritating substances such as tints or chemicals, 236

Dermatology, a branch of medical science that deals with the study of the skin, 222

Dermis, second or inner layer of the skin; also known as the *derma*, *corium*, *cutis*, or *true skin*, 223, 224–225, 228

Design elements in haircutting, 395–397

Design line, usually the perimeter line of a haircut, 400

Design texture, 397

Desincrustation, the process of deep pore cleansing using an electrode and acid-based solution to create a chemical reaction that helps to emulsify or liquefy sebum and waste from the skin, 212, 317

Detachable-blade clippers, 115

Developer, also known as *oxidizing agents* or *catalysts*; when mixed with an oxidation haircolor, supplies the necessary oxygen gas to develop color molecules and create a change in hair color, 654–656

Diagnosis, determination of the nature of a disease from its symptoms and/or diagnostic tests; federal regulations prohibit salon professionals from performing a diagnosis, 68

Diagonal lines, lines positioned between horizontal and vertical lines, 396

Diamond-shaped face, 389

Diffused drying, 419, 478

Digestive system, the mouth, stomach, intestines, and salivary and gastric glands that change food into nutrients and wastes, 148

Diplococci, spherical bacteria that grow in pairs and cause diseases such as pneumonia, 65

Diplomacy, 36

Direct application, 324

Direct current (DC), constant current that travels in one direction only and produces a chemical reaction, 207–208

Direct surface application, high-frequency current performed with the mushroom- or rake-shaped electrodes for its calming and germicidal effect on the skin, 316

Direct transmission, transmission of blood or body fluids through touching (including shaking hands), kissing, coughing, sneezing, and talking, 64

Directional terms, 397–398

Discolored, a condition in which the nails turn a variety of colors such as yellow, blue, blue-gray, green, red, or purple; can be caused by poor blood circulation, a heart condition, topical or oral medications, or a systemic disorder, 784

Discretion, 36

Disease, an abnormal condition of all or part of the body, or its systems or organs, that makes the body incapable of carrying on normal function, 60, 64, 68

Disinfectants, chemical products approved by the EPA designed to destroy most bacteria (excluding spores), fungi, and viruses on surfaces, 60–61

for containers, tools, and work surfaces, 77

nonelectrical tools and implements, 86–88

proper use of, 74–75

safety with, 76

types of, 75–76

Disinfection (disinfecting), a chemical process that uses specific products to destroy harmful organisms (except bacterial spores) on environmental surfaces, 56, 63

Disulfide bonds, also known as a *sulfur bond*; a type of chemical cross bond found in the hair cortex, 251, 252, 253

Double flat wrap mini-procedure, 592

Double-process haircoloring, 669–670, 689–693

Double-process haircoloring, a two-step combination of lightening and haircoloring, 663

Drapes, 127

Draping, the term used to describe the covering of the client's clothing with a cape for their protection, 277–279, 286–289

Dress codes, 43

Dressing for success, 42–43

Dry or damaged hair, 300, 603

Dry or powder shampoos, shampoos that cleanse the hair without water, 193

Dry peroxides, 656

Dry scalp and hair, treatment for, 283, 298

Dry skin, 320, 325

Duct glands, 166

Ductless glands, 166

Dye removers, products used to strip built-up color from the hair, 660

Dyschromias, abnormal skin colorations, 237

E

- Ear muscles, 153
- Ear size and placement, 391
- Eczema**, inflammatory skin condition characterized by painful itching; dry or moist lesion forms, 236
- Effective communication**, the act of successfully sharing information between two people (or groups of people) so that the information is understood, 49–50
- Efferent nerves, 158
- Efficacy**, the ability of a product to produce the intended effect; on a disinfectant label, it indicates specific pathogens destroyed or disabled when used properly, 74
- Efficiency, 751
- Effleurage**, light, continuous stroking movement applied with the fingers (digital) or the palms (palmar) in a slow, rhythmic manner, 311
- Elasticity. *See* Hair elasticity; Skin elasticity
- Elastin**, protein base similar to collagen that forms elastic tissue, 226
- Electric current**, the flow of electricity along a conductor, 207–209
- Electric latherizer, 128–129
- Electric massager**, massaging unit that attaches to the barber's hand to impart vibrating massage movements to the skin surface, 130, 285, 313, 328
- Electric stove, 130
- Electric wire, 207
- Electrical measurements, 208–209
- Electricity**, a form of energy that produces physical, magnetic, chemical, or thermal effects when in motion, 206
- electric current, types of, 207–209
 - electrical equipment safety, 209–211
 - guidelines for safe use, 210–211
 - light energy and light therapy, 213–217
 - overview of, 206–209
 - safe work practices and safety precautions, 81
 - safety precautions, 319–320
 - terminology, 211–213
- Electrode**, an applicator used to direct electric current from a machine to the skin, 211, 315
- Electromagnetic spectrum**, the range of all forms of electromagnetic radiation, 213–214
- Electrotherapy**, electronic scalp and facial treatments, 211, 217, 315–320
- Electrotherapy equipment, 315–320
- Electrotherapy machines, 131–132
- Element**, is the simplest form of chemical matter and contains only one type of atom, 179–180
- Elemental molecules**, molecules consisting of two or more atoms of the same element in fixed proportions, for example, oxygen (O₂), 180
- Elevation**, angle or degree at which a subsection of hair is held, or elevated, from the head when cutting; also referred to as projection, 398–402
- Eleventh cranial nerve**, spinal nerve branch that affects the muscles of the neck and back, 160, 308
- Emotional control, 48
- Emotional disturbances, 249
- Emotional stability, 36
- Employee**, employment classification in which the employer withholds certain taxes and has a high level of control, 738, 741
- Employee contracts, 729–730
- Employee evaluation, 742
- Employee status, 738
- Employment
- application for, 730
 - barbershop visit, 723–724
 - employee contracts, 729–730
 - field research, 722
 - getting the job you want, 716
 - inventory of characteristics and skills, 715
 - job interview, 724–730
 - opportunities, 716
 - portfolio, 719–721
 - preparing for, 714–724
 - resume development, 716–719
 - targeting the establishment, 721–722
- Employment classifications, 738–739, 741
- Employment portfolio**, a collection, usually bound, of photos and documents that reflect your skills, accomplishments, and abilities in your chosen career field, 719–721
- Emulsions**, mixtures of two or more immiscible substances united with the aid of a binder or emulsifier, 185–186
- End bonds**, also known as *peptide bonds*; chemical bonds that join amino acids end to end, 251
- End papers, 589–592
- End wraps**, end paper; absorbent papers used to protect and control the ends of the hair during perming services, 589–592
- Endocrine glands**, also known as *ductless glands*; release hormonal secretions directly into the bloodstream, for example, thyroid and pituitary glands, 166, 249
- Endocrine system**, group of specialized glands that affect growth, development, sexual function, and general health, 148, 166–167
- Endothermic reaction**, a chemical reaction that requires the absorption of energy or heat from an external source for the reaction to actually occur, 183
- Endothermic waves**, perm activated by an outside heat source, usually a hood-type dryer, 596
- Enthusiasm, 716
- Environmental Protection Agency (EPA), 57, 60–61
- Envision**, the ability to picture or see something in your mind, 386
- Epicranial aponeurosis**, a tendon that connects the occipitalis and frontalis muscles, 152
- Epicranius**, also known as *occipitofrontalis* broad muscle that covers the top of the skull and consists of the occipitalis and frontalis, 152
- Epidermis**, outermost layer of the skin; also called the *cuticle* or *scarf skin*, 223–224, 228
- Epilators, 198
- Epithelial tissue**, a protective covering on body surfaces, such as skin, mucous membranes, the tissue inside the mouth, the lining of the heart, digestive and respiratory organs, and the glands, 147
- Eponychium**, living skin at the base of the nail plate and covering the matrix area, 782–784, 804
- Equal Employment Opportunity Commission (EEOC), 772
- Equipment, safe work practices and safety precautions, 82
- Ergonomics**, the science of designing the workplace as well as its equipment and tools to make specific body movements more comfortable, efficient, and safe, 44–47
- Essay questions, 712
- Ethical standards, 34–36
- Ethics**, the moral principles by which we live and work, 34
- Ethmoid bone**, a light, spongy bone between the eye sockets forming part of the nasal cavities, 149
- Eumelanin**, melanin that gives brown and black color to hair, 253, 644
- Excoriation**, skin sore or abrasion caused by scratching or scraping, 232
- Excretion, 229
- Excretory system**, group of organs including the kidneys, liver, skin, large intestine, and lungs that purify the body by the elimination of waste matter, 148
- Exercise, 42
- Exits, 84
- Exocrine glands**, also known as *duct glands*; produce a substance that travels through small, tubelike ducts; for example, sweat and oil glands of the skin, 166
- Exothermic perm activator, 182
- Exothermic reaction**, the release of energy in the form of heat through certain chemical reactions, 183

Exothermic waves, perms that create an exothermic chemical reaction that heats the solution and speeds up processing, 597

Exposure incident, contact with nonintact (broken) skin, blood, body fluid, or other potentially infectious materials, which is the result of the performance of an employee's duties, 68, 80, 95–97

Extensors, muscles that straighten the wrist, hand, and fingers to form a straight line, 156

External carotid artery, artery that supplies blood to the anterior parts of the scalp, face, neck, and side of the head, 163

External jugular vein, vein located at the side of the neck that carries blood returning to the heart from the head, face, and neck, 165

External maxillary artery, 164, 309

Extremely curly hair, 254, 255

Eye muscles, 154

F

Face arteries, 163–164

Face veins, 164–165

Facebook, 751

Facial appliances, 313–315

Facial artery, artery that supplies blood to the lower region of the face, mouth, and nose, 164

Facial bones, 149–150

Facial equipment

electrotherapy equipment, 315–320

facial appliances, 313–315

purpose of, 313–320

Facial hair replacement systems, 504–505

Facial massage and treatments, 326–328
arteries and veins affected by facial massage, 309

basic facial, 324

basic facial procedure, 331–333

brush machine, 329

cranial nerves affected by facial massage, 308

for dry skin, 325

facial steamer, 330

facial treatments, 320–325

guidelines for, 325

muscles affected by facial massage, 305–308

for oily skin, 325

purpose of facial equipment, 313–320

skin care products, 322–324

skin types, 320–324

subdermal systems, 304–309

theory of massage, 309–313

vibratory facial, 324

Facial muscles, 153

Facial nerve, 159–160, 308

Facial shape, oval, round, inverted triangular, square, oblong, diamond, and pear-shaped are the seven facial shapes, 387–389

Facial skeleton, two nasal bones, two lacrimal bones, two zygomatic bones, two maxillae, the mandible, two turbinal bones, two palatine bones, and the vomer, 149

Facial steamer, an electrical appliance that produces and projects moist, uniform steam for softening and cleansing purposes, 314, 330

Facial-hair design, 349, 350, 351

Fade style, 413

Fade with star design, 465–468

Fatty alcohols, 195

Federal agencies, 57–61

Field research, 722

Fifth cranial nerve, chief sensory nerve of the face; controls chewing, 158, 160, 308

Fillers, preparations designed to equalize porosity and/or deposit a base color in one application, 660–661

Finasteride, an oral medication prescribed for men only to stimulate hair growth, 260, 498

Fine hair, 266, 643–644

Finger brace, 109

Finger grip and tang, 109, 110

Finger styling, 417

Finger waving, 418, 540

Fingers-and-razor cutting, cutting technique performed by holding the hair section between the fingers and cutting either from the top to the bottom of the section or from one side to the other; also known as *freehand slicing* with a razor, 408

Fingers-and-shear, technique used to cut hair by holding the hair into a position to be cut, 403, 422–423

Finish work, 415–417

Fire extinguishers, 84

First-time-over shave, first part of the standard shave consisting of shaving the 14 areas of the face; followed by the second-timeover shave to remove residual missed or rough spots, 340, 348

Fissure, a crack in the skin that penetrates to the dermis, 233

Fixtures, 82

Flagella, slender, hairlike extensions used by bacilli and spirilla for locomotion (moving about). May also be referred to as *cilia*, 65

Flap surgery, a surgical technique that involves the removal of a bald scalp area and the attachment of a flap of hair-bearing skin, 499

Flat irons, 130, 543, 573–574

Flat top, 412, 450–452

Flexor, extensor muscle of the wrist involved in flexing the wrist, 156

Floor plans, 769

Foil frost, highlighting technique using foil, 663

Foil technique, coloring or highlighting technique using foil to apply product to specific hair sections, 663

Follicle, tubelike depression in the skin that contains the hair root, 248–249

Folliculitis barbae, also known as *barber's itch*; bacterial infection of the hair follicles with inflamed pustules in the bearded areas of the face and neck; may have hairs growing through the pustule, 264

Food and Drug Administration (FDA), 79, 256

Forged shears, shears that have been made by working heated metal into a finished shape through the processes of hammering or compression, 108–109

Form, the outline or shape of a design, 396

Form 1099-MISC, 739

Form W-2, 738

Formaldehyde, 200

Fragilitas crinium, technical term for brittle hair, 261

Franchise ownership, 762

Francis I of France, 10

Free edge, part of the nail plate that extends over the tip of the finger, 63, 79, 94, 782–785, 787, 789–791, 804–805

Freeform blowdrying, 418, 477

Free-form technique, also known as balyage or baliaage; the painting of a lightener on clean, styled hair, 663

Freehand, razor position and stroke used in 6 of the 14 shaving areas: nos. 1, 3, 4, 8, 11, and 12, 342–343

Freehand clipper cutting, generally interpreted to mean that guards are not used in the cutting process, 405–407

Freehand notching, 538

Freehand position and stroke, 344–345, 355

Freehand shear cutting, cutting with shears without the use of fingers or a comb to control the hair, 405

Freehand slicing, method of removing bulk from a hair section with the shears, 408

Fresheners, skin tonics with the lowest alcohol content of 0 to 4 percent; usually designed for dry, mature, and sensitive skin types, 197, 322

Friction, deep rubbing movement requiring pressure on the skin with the fingers or palm while moving the hand over an underlying structure, 311

Frontal artery, artery that supplies blood to the forehead and upper eyelids, 164

Frontal bone, bone that forms the forehead, 149

Frontalis, anterior or front portion of the epicranium; muscle of the scalp, 152

Full head bonding, the process of attaching a hair replacement system to all areas of the head with an adhesive bonding agent, 504, 514–516

Full wigs, 505

Full-charge bookkeeper, 766

Fundus, 227

Fungi, single-celled organisms that grow in irregular masses and include molds, mildews, and yeasts; they can produce contagious diseases such as ringworm, 70–71

Fungicidal, capable of destroying molds and fungi, 63

Furuncle, an acute bacterial infection of a hair follicle, producing constant pain; also known as a boil, 264, 265

Fuse, device that prevents excessive current from passing through a circuit, 209

G

Galvani, Luigi, 212

Galvanic current, constant and a DC, having a positive and negative pole, that produces chemical changes in tissues and body fluids, 212, 316–317

Galvanic machine, 132, 316

Game plan, the conscious act of planning your life, instead of just letting things happen, 26

Gamma rays, 214

Gel masks, 323

General circulation. *See* Systemic circulation

Germs, 64

Glands, specialized organs varying in size and function that have the ability to remove certain elements from the blood and to convert them into new compounds, 166, 227–228

Globally Harmonized System of Classification and Labeling of Chemicals System (GHS), 58

Glycerin, 200

Glyceryl monothioglycolate, main active ingredient in true acid and acid-balanced waving lotions, 596, 599

Goals, a set of benchmarks that, once achieved, help you to realize your mission and your vision, 757

Goals setting, the identification of long-term and short-term goals that helps you decide what you want out of your life, 28–30

Good manners, 49

Goose bumps, 249

Graduated cut, graduated, wedge, or stacked shape at the perimeter of a haircut, usually cut at 45 degrees, 533, 548–553

Grain, 397–398, 406

Grain change, 340

Granular layer, 224

Gray hair, 257, 648–649, 672–673

Greater auricular nerve, nerve at the sides of the neck affecting the face, ears, and neck, 160

Greater occipital nerve, nerve located at the back of the head, affecting the scalp, 160

Greece, 8

Greeting clients, 48

Ground fault interrupter (GFI), a device that senses imbalances in an electric current, 81, 210

Grounding, 209–210

Grounding pin, 210

Growth phase, 256

Guanidine hydroxide relaxers, 606

Guarantee, 740

Guards, plastic or hard rubber comb attachments that fit over clipper blades to minimize the amount of hair being cut with the clippers; or a metal shield applied over a haircutting razor for protection, 117

Guide, section of hair, located at either the perimeter or the interior of the cut, that determines the length the hair will be cut to; also referred to as a guideline; usually the first section that is cut to create a shape, 400

H

Hackling, process used to comb through the hair strands to separate them, 499

Hadrian, Emperor, 8

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scalp treatment variations, 297–300

shampoo and shampoo massage manipulations, 290–292

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shampoo service, understanding, 279–281

shampoo variations, 293

Hair bulb, club-shaped structure that forms the lower part of the hair root, 249

Hair clippings, 132

Hair color crayons, 674

Hair conditioners, products designed to moisturize the hair or restore some of the hair's oils or proteins, 277

Hair cortex, 251–253

Hair cuticle, 249

Hair density, the amount of hair per square inch of scalp, 267, 281, 584, 643

Hair elasticity, the ability of the hair to stretch and return to its original length, 268, 281, 583–584, 643

Hair growth, 255–257

Hair growth direction, 406

Hair growth patterns, 584–585

Hair lightening, the chemical process of diffusing natural or artificial pigment from the hair, 642, 656–657. *See also* Haircoloring

- contribution of underlying pigment, 658, 642, 656–657
- fillers, 660–661
- lightener retouch, 670–671
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- selection of lighteners, 670
- skin stain removers, 661
- special effects of, 672
- tint or dye removers, 660
- toners, 659–660
- types of, 657–658

Hair locking, the process that occurs when coiled hair is allowed to develop in its natural state without the use of combs, heat, or chemicals, 420–421

Hair loss, 257–260

Hair maintenance, 43

Hair masks, 195

Hair molding, styling method that uses the head as tool to set the hair in a straight position, 541

Hair porosity, the ability of the hair to absorb moisture, 267–268, 281, 581–582, 643, 669

Hair pressing, temporarily straightening hair by using heated pressing combs, 542–543, 571–572

Hair replacement system, formerly called a *hairpiece*; also known as a *hair solution*, 494

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Hair root, the part of the hair that is encased in the hair follicle, 248–249

Hair shaft, the part of the hair that extends beyond the skin, 248, 249–250

Hair solution, any small wig used to cover the top or crown of the head and integrated with the natural hair, 494

Hair spray, 198

Hair stream, hair that flows in the same direction, 257

Hair texture, measures the diameter of a hair strand; coarse, medium, fine, 266, 277, 281, 583, 643

Hair thinning and texturizing, 410

Hair tonic treatments, 285

Hair tonics, grooming aids designed to stimulate the surface circulation of the scalp, remove loose dandruff, or to impart manageability, shine, and control to the hair, 196–197

Hair transplant, 260

Hair transplantation, any form of hair restoration that involves the surgical removal and relocation of hair, including scalp reduction and flap surgery, 498

Hair types, matching products to, 196, 277

Hair wrapping, method where the hair is wrapped around the head for drying and styling purposes, 540, 565–567

Haircolor mousses and gels, 651

Haircolor record card, 664

Haircolor sprays, 651

Haircoloring, industry-coined term referring to artificial haircolor products and services; the addition of color on or into the hair shaft, 254, 642. *See also* Hair lightening

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Haircutting and styling. *See* Men's haircutting and styling; Women's haircutting and styling

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Haircutting shears, 20, 108–112

Hairdressings, 198

Hairpiece, 494

Hairstyling, the art of arranging the hair in a particular style that is appropriately suited to the cut, 402. *See also* Men's haircutting and styling; Women's haircutting and styling

Half off-base, position of a curl one-half off its base; provides medium volume and movement, 542

Half-base, 542

Half-stem, 542

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Hand Massager, 285

Hand muscles, 156–157

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Hangnails, condition in which the cuticle splits around the nail, 784, 789

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Head veins, 164–165

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Heart, muscular, cone-shaped organ that keeps blood moving through the circulatory system, 161

Heat receptor, 226

Heat regulation, 229

Heavier taper-blending, 409

Heel, 115

Helix, the spiral shape of a coiled protein created when polypeptide chains intertwine, 251

Henna, 654

Henry VIII, King, 11, 12

Hepatitis, a bloodborne virus that causes disease and can damage the liver, 69–70

Herpes simplex 1, fever blister or cold sore; a recurring viral infection, 236–237

Herpes simplex II, 237

High and tight cuts, 413

High porosity, 267–268, 643

High-frequency machine, a machine that produces a high rate of oscillation or Tesla current for the purpose of stimulating scalp, facial, and body tissues, 131, 315–316

Highlighting, coloring or lightening some strands of hair lighter than the natural color, 663

Highlighting colors, 646

Hirsuties, 261

Histology, the study of the minute structure of the various tissues and organs that make up the entire body of an organism, 145

HIV. *See* Human immunodeficiency virus (HIV)

Hone, a sharpening block manufactured from rock or synthetic materials and used to create a cutting edge on conventional straight razors, 122–124, 133–134

Honeycomb ringworm, 263

Horizontal lines, lines parallel to the horizon, 395

Hormones, secretions that stimulate functional activity or other secretions in the body, for example, insulin, adrenaline, and estrogen, 167

Horny layer, 223

Hospital disinfectants, disinfectants that are effective for cleaning blood and body fluids on nonporous surfaces, 60

Hospital grade, 58

Hot towel cabinet, 128, 314

Hue, the basic name of a color, 646

Human immunodeficiency virus (HIV), abbreviated HIV; virus that causes HIV disease and acquired immune deficiency syndrome (AIDS), 70

Human relations, interactions and relationships between two or more people, 47–50

Humectants, 193, 195

Humerus, uppermost and largest bone in the arm, 151

Hunchback, 46

Hydrogen bond, weak, physical, cross-link side bond that is easily broken by water or heat, 251–252

Hydrogen peroxide, 79, 200, 655

Hydrogen peroxide developer, 654–656

Hydroxide, 610

Hydroxide relaxers, very strong alkalis with a pH over 13; the hydroxide ion is the active ingredient in all hydroxide relaxers, 587, 604–610, 626–631

Hygiene, 41

Hyoid bone, U-shaped bone at the base of the tongue at the front part of the throat, 150

Hyperhidrosis, excessive perspiration or sweating, 236

Hyperpigmentation, darker than normal pigmentation, appearing as dark splotches, 237

Hypertrichosis, a condition of abnormal hair growth, 261

Hypertrophy, abnormal skin growth, 238–239, 266

Hyponychium, slightly thickened layer of skin between the fingertip and free edge of the nail plate, 782–783

Hypopigmentation, the absence of pigment, resulting in light or white splotches, 237

I

Image-building basics, 42–44

Immune system, 148. *See also* **Lymphatic/immune system**

Immunity, the ability of the body to destroy and resist infection; immunity against disease can be either natural or acquired and is a sign of good health, 71

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- barber chair, 126–127
- clippers and outliners, 114–117
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- drapes and capes, 127
- equipment and supplies, 126–132
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- holding a straight razor, 120–126
- holding clippers and trimmers, 117–118
- learning about, 105
- shears, holding, 112–113
- straight razors, 119–120
- towels and linens, 127–128

Inactive stage, also known as *spore-forming stage*; the ability of some bacteria to form a protective coating around themselves to protect them from harsh environments during the inactive stage, 66

Inclined method, 279

Income, reporting, 743–746

Incoming telephone calls, 774

Independent contractor, A form of self-employment and tax designation with specific responsibilities for book-keeping, taxes, insurances, and so on, 738–739, 741

Indirect application, high-frequency current administered with the client holding the wire glass electrode between both hands, 316, 324

Indirect transmission, transmission of blood or body fluids through contact with an intermediate contaminated object such as a razor, extractor, nipper, or an environmental surface, 64

Infection, the invasion of body tissues by disease-causing pathogens, 62, 64

Infection control, the methods used to eliminate or reduce the transmission of infectious organisms, 56–57

- bacteria, 63–67
- biofilms, 69
- bloodborne pathogens, 69–70
- cleaning, 72–73
- cleaning towels, linens, and capes, 77–78
- clippers and outliners, 89–92
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- safe work practices and safety precautions, 80–84
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- standard precautions, 80
- strops and leather holsters, 78
- viruses, 67–69
- waterless hand sanitizers, 79

Infectious, caused by or capable of being transmitted by infection, 62–63

Infectious disease, disease caused by pathogenic (harmful) microorganisms that enter the body; an infectious disease may or may not be spread from one person to another person, 63, 68

Inferior labial artery, artery that supplies blood to the lower lip, 164

Inflammation, a condition in which the body reacts to injury, irritation, or infection, characterized by redness, heat, pain, and swelling, 66, 68

Infraorbital artery, artery that supplies blood to the eye muscles, 164

Infraorbital nerve, nerve that affects the skin of the lower eyelid, side of the nose, upper lip, and mouth, 159

Infrared light, invisible rays with long wavelengths and deep penetration; produce the most heat of any therapeutic light, 215–216, 217

Infrared-ray lamps, 318

Infratrochlear nerve, nerve that affects the membrane and skin of the nose, 159

Inorganic chemistry, is the study of substances that do not contain carbon but may contain hydrogen, 179

Insertion, the part of the muscle that moves and is furthest from the skeleton, 151, 152

Instant conditioners, conditioners that typically remain on the hair from 1 to 5 minutes and are rinsed out; also known as *rinse-out conditioners*, 194

Insulator, substance that does not easily transfer electricity, 207

Insurance, guarantees protection against financial loss from malpractice, property liability, fire, burglary and theft, and business interruption, 760

Integrity, 35, 716

Integumentary system, the skin and its appendages—glands, sensory receptors, nails, and hair, 148, 165–166

Interests self-test, 28

Internal carotid artery, artery that supplies blood to the brain, eyes, eyelids, forehead, nose, and ear, 163

Internal jugular vein, vein located at the side of the neck; collects blood from the brain and parts of the face and neck, 165

Interstitial fluid, blood plasma found in the spaces between tissue cells, 165

Inventory of Personal Characteristics and Technical Skills, 714, 715

Inverted triangular shaped face, 388

Invisible light, 215–216

Ion, an atom or molecule that carries an electrical charge, 187–188

Ionization, the separating of a substance into ions, 187–188

Iontophoresis, the process of using galvanic current to enable ion-containing water-soluble solutions to penetrate the skin, 212, 317

Irritant contact dermatitis, abbreviated ICD; also known as *dermatitis venenata*; occurs when irritating substances temporarily damage the epidermis, 236

Isopropyl alcohol, 79

Ivy dermatitis, a skin inflammation caused by exposure to poison ivy, poison oak, or poison sumac, 237

J

Jewelry, 82

Job description, document that outlines all the duties and responsibilities of a particular position in a barbershop, 737–738

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wardrobe, 726

Joint, the connection between two or more bones of the skeleton, 149

Journeyman barber groups, organizations formed to represent barber employees prior to the establishment of barber licensing laws, 13–14

Journeyman Barbers, Hairdressers, Cosmetologists, and Proprietors' International Union of America, 14

Journeyman Barbers' International Union of America, 14

K

Keloid, thick scar resulting from excessive tissue growth, 233

Keratin, the protein of which hair is made, 223, 248, 250

Keratinization, process by which protein cells mature within the follicle to form hair, 250

Keratoma, technical name for a callus, caused by pressure or friction, 238

Knotting, 500–501

L

Lace-front, popular hair solution style used for off-the-face styles, 504, 517–518

Lacrimal bones, small bones located in the wall of the eye sockets, 150

Lanthionization, process by which hydroxide relaxers permanently straighten hair; lanthionization breaks the hair's disulfide bonds during processing and converts them to lanthionine bonds when the relaxer is rinsed from the hair, 587

Lanugo, vellus hair, 255

Laser, acronym for light amplification by stimulated emission of radiation, 216

Lather brushes, 129

Lather mugs, 129

Latherizers, 128–129

Latissimus dorsi, a large, flat, triangular muscle covering the lower back; helps extend the arm away from the body and rotate the shoulder, 155

Laws, vs. rules, 62

Laws of color, a system for understanding color relationships, 644

Layers, graduated effect achieved by cutting the hair with elevation or overdirection; the hair is cut at higher elevations, usually 90 degrees or above, which removes weight, 401

Learning tools, 31–34

Leather holsters, 78

Leave-in conditioners, conditioners and thermal protectors that can be left in the hair without rinsing, 195

Left-handed barbers, shaving movements for, 341–342

Left-handed shears, 113

Lentigines, technical name for freckles, 238

Lesion, a structural change in the tissues caused by injury or disease, 230

Leukoderma, skin disorder characterized by abnormal white patches, 238

Levator anguli oris muscle, muscle that raises the angle of the mouth and draws it inward, 154

Levator labii superioris muscle, muscle surrounding the upper lip, 154

Levator palpebrae superioris muscle, a thin muscle that controls the eyelid, 154

Level, unit of measurement to identify the lightness or darkness of a color, 646–648

Level system, system used to analyze the lightness or darkness of a hair color or color product, 647

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Lifelong learning, 736

Light energy, 213–217

Light taper-blending, 409

Light therapy, also known as *phototherapy*, is the application of specific wavelengths of the light spectrum to the skin for the treatment of specific disorders or conditions, 213–217, 317–318

Light therapy machines, 131–132

Light waves, 214

Light-emitting diode (LED), medical device used to reduce acne, increase skin circulation, and improve collagen content in the skin, 216–217

Lightener retouch, 670–671

Lighteners, chemical compounds that lighten hair by dispersing and diffusing natural pigment, 656–661, 670–671. *See also* Hair lightening

Line of demarcation, a visible line separating colored hair from new growth, 653

Linens, 77–78, 127–128

Lines, 395–396

Liquid eyebrow and eyelash tints, 674

Liquid hydrogen peroxides, 655

Liquid-dry shampoos, liquid shampoos that evaporate quickly and do not require rinsing, 192

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Liver spots, 238

LLC (Limited Liability Company), 761

Local business referrals, 751

Local infection, an infection, such as a pimple or abscess, that is confined to a particular part of the body and appears as a lesion containing pus, 67

Location of barbershop, 758

Locks, 420–421, 479–482, 486–488

Long stem, 542

Long-layered cut, hair is cut at a 180-degree elevation to create short layers at the top and increasingly longer layers at the perimeter, 534–535, 559–564

Lotion wrap, permanent waving wrapping technique in which the waving solution is applied to the section before rodding, 594–595

Louis XIV, King, 10

Low porosity, 267–268, 643

Low-level laser therapy, also known as *low-level light therapy* or *cold-laser therapy*; uses low levels of red and near infrared light energy to stimulate hair growth, 216, 305

Low-light laser therapy, 498

Lowlighting, coloring some strands of hair darker than the natural hair color, 663

Low-pH waves, 598

Loyalty, 737

Lunula, half-moon shape at the base of the nail, 782–783, 785, 789

Lymph, colorless, watery fluid that circulates in the lymphatic system; carries waste and impurities from cells, 165

Lymph capillaries, blind-end tubes occurring individually or in clusters that are the origin of lymphatic vessels, 165

Lymph nodes, glandlike structures found inside lymphatic vessels that filter lymph, 165

Lymphatic/immune system, consists of lymph, lymph nodes, the thymus gland, the spleen, and lymph vessels that act as an aid to the blood system, 165

M

Macule, spot or discoloration of the skin, such as a freckle, 231

Magnetic motor clippers, 115–116

Makeup, 43

Malar bones, 150

Malassezia, fungus that causes dandruff, 261

Male pattern baldness, 258, 259

Malignant melanoma, most severe form of skin cancer, 239

Malpighian layer, 224

Mandible, lower jawbone, 150

Mandibular nerve, branch of the fifth cranial nerve that affects the muscles of the chin, lower lip, and external ear, 159

Mannerisms, 49

Marcel iron, 130, 131

Marginal mandibular nerve, affects the muscles of the chin and lower lip, 160

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Masks, 198, 322–323

Massage, theory of

massage manipulations, 310–313

physiological effects of, 309–310

Massage creams, 198, 323

Massage manipulations, 310–313

Masseter, one of the jaw muscles used in chewing, 153

Master barber groups, employer organizations formed prior to the establishment of barber licensing laws, 13

Mastication muscles, 153

Matching questions, 712

Material Safety Data Sheet (MSDS), 58

Matrix, area where the nail is formed; produces cells that create the nail plate, 782–785

Matter, is defined as anything that occupies space (volume) and has mass (weight), 179–182

Maxillae, bones of the upper jaw, 150

Maxillary nerve, supplies impulses to the upper part of the face, 159

Meal planning, 41

Mechanical exfoliants, 322

Medicated conditioners, conditioners formulated to control minor dandruff and scalp conditions, 195

Medicated lotions, 198

Medicated shampoos, shampoos containing medicinal agents for controlling dandruff and other scalp conditions, 193–194

Medium elevation, 398

Medium hair, 266, 643–644

Medium press, 542

Medulla, innermost or center layer of the hair shaft, 250

Melanin, coloring matter or pigment of the skin; found in the stratum germinativum of the epidermis and in the papillary layers of the dermis, 224, 226–227; **tiny grains of pigment in the cortex that give natural color to the hair,** 644, 654–655, 657

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techniques, 402–411

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trimming eyebrows, 416

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vs. women's, 531

Mental nerve, nerve that affects the skin of the lower lip and chin, 159

Mentalis muscle, muscle that elevates the lower lip and raises and wrinkles the skin of the chin, 154

Meryma'at, Egyptian barber commemorated with a statue, 7

Metacarpals, the bones of the palm, consisting of five slender bones between the carpus and the phalanges, 151

Metallic dye, 603, 654

Metallic or progressive dyes, 674

Metallic salts, testing for, 675

Methicillin-resistant staphylococcus aureus (MRSA), Abbreviated MRSA; a type of infectious bacteria that is highly resistant to conventional treatments such as antibiotics, 67

Microcurrent, a type of galvanic treatment that uses a very low level of electrical current for different applications in skin care, 212–213, 317

Microdermabrasion, a form of mechanical exfoliation that involves spraying aluminum oxide or other microcrystals across the skin's surface to exfoliate dead cells, 317

Microorganism, any organism of microscopic or submicroscopic size, 63, 64

- Microwaves, 214
- Middle temporal artery**, artery that supplies blood to the temples, 164
- Mildew**, a type of fungus that affects plants or grows on inanimate objects but does not cause human infections in the barbershop, 70
- Milia**, technical name for milk spots; small, benign, whitish bumps that occur when dead skin is trapped in the surface of the skin; commonly seen in infants, 234
- Miliaria rubra**, also known as prickly heat, is an acute inflammatory disorder of the sweat glands, characterized by the eruption of small red vesicles accompanied by burning, itching skin, 236
- Milliamperes, 208
- Mind mapping**, a graphic representation of an idea or problem that helps organize one's thoughts, 33–34
- Mineral dyes, 654
- Minoxidil**, topical medication used to promote hair growth or reduce hair loss, 260, 498
- Mission statement**, a statement that establishes the purpose and values for which an individual or institution lives and works by. It provides a sense of direction by defining guiding principles and clarifying goals, as well as how an organization operates, 27–28, 757, 763
- Mitosis**, cells dividing into two new cells (daughter cells), 146
- Mixed bristle brush, 108
- Mixed hair, 500
- Mnemonic**, any memorization device that helps a person recall information, 32
- Modalities**, currents used in electric facial and scalp treatments, 211–213
- Moisturizers, 195
- Moisturizing creams, 198
- Moisturizing shampoo, 193
- Mole**, small brownish spot on the skin, 238
- Molecules**, two or more atoms joined chemically, 180
- Moler, A. B.**, in 1893 established America's first barber school in Chicago, Illinois, 18
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give yourself a raise, 745
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seek professional advice, 745–746
- Monilethrix**, technical term for beaded hair, 261
- Motility**, self-movement, 65
- Motivation, 716
- Motor nerve fibers**, nerve fibers distributed to the arrector pili muscles, which are attached to the hair follicles, 225
- Motor nerves**, nerves that carry impulses from the brain to the muscles, 158
- Motor point**, a point on the skin, over a muscle, where pressure or stimulation will cause contraction of that muscle, 310
- Mouth muscles, 154–155
- Moving blade, 109
- Moving point, 109
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- Multiple quats, 75
- Multiuse**, also known as *reusable*; items that can be cleaned, disinfected, and used on more than one person, even if the item is accidentally exposed to blood or body fluid, 73
- Multiuse products, 78
- Muscle tissue**, contracts and moves various parts of the body, 147
- Muscular system**, body system that covers, shapes, and supports the skeletal tissue, 148, 151
arm and shoulder muscles, 155–156
hand muscles, 156–157
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of the scalp, face, and neck, 152–155, 307–308
stimulation of, 152
structure of, 151
- Mustache design, 350
- Mustaches, 278, 374, 674, 699–701
- Myology**, study of the structure, function, and diseases of the muscles, 151
- ## N
- Nail**, an appendage of the skin; horny protective plate at the end of the finger or toe, 63, 67–68, 73, 79, 82, 93–94, 124, 782
- Nail bed**, portion of the skin on which the nail plate rests, 782
- Nail care, 44
- Nail cuticle, 232
- Nail folds**, folds of normal skin around the nail plate, 94, 783
- Nail grooves**, slits or furrows on the sides of the nail, 783
- Nail plate**, horny plate resting on and attached to the nail bed, 79, 782
- Nasal bones**, bones that form the bridge of the nose, 150
- Nasal nerve**, nerve that affects the point and lower sides of the nose, 159
- National Association of Barber Boards of America (NABBA)**, 15
- National Association of Barber Schools, 15
- National Association of State Board of Barber Examiners, 15
- National Barber Museum, 17
- Natural bristle brush, 108
- Natural dry styling, 540
- Natural drying, 417
- Natural hair color, 644, 674
- Natural haircolors, 654
- Natural hones, 123
- Natural immunity**, immunity that is partly inherited and partly developed through healthy living, 71
- Natural part, 399
- Neck arteries, 163–164
- Neck bones, 150
- Neck duster, 132
- Neck lengths, 390–391
- Neck muscles, 153
- Neck shave**, shaving the areas behind the ears down the sides of the neck, and at the back neckline, 349, 372–373, 416
- Neck veins, 164–165
- Negative space, 397
- Nerve cell, 158
- Nerve tissue**, type of tissue that carries messages to and from the brain and controls and coordinates all bodily functions, 147
- Nerves**, whitish cords made up of bundles of nerve fibers held together by connective tissue, through which impulses are transmitted, 157, 158, 225–226, 305
- Nervous system, 148, 157–160
- Neurology, 157
- Neuron**, also known as *nerve cell*; the primary structural unit of the nervous system, consisting of the cell body, nucleus, dendrites, and the axon, 158
- Neutral fillers, 661
- Neutral oil lighteners, 658
- Neutralization**, process of stopping the action of a permanent wave solution and hardening the hair in its new form by the application of a chemical solution called the neutralizer, 601–602
- Neutralizing shampoos**, helps to rebalance and restore the pH level of the hair by neutralizing any alkali or unwanted residues that remain in the hair after chemically relaxing the hair, 194
- Nevus**, technical name for a birthmark, 238
- 90-degree elevation, 398–399
- No-base relaxers**, relaxers that do not require application of a protective base, 607
- Nodule**, a solid bump larger than 0.4 inches (1 cm) that can be easily felt, 231
- No-fragrance policy, 42
- No-lift, deposit-only haircolors, 651–652
- No-mix/no-lye relaxers, 606
- Nonionics, 192
- Nonoxidation color, 650–651
- Nonpathogenic**, harmless microorganisms that may perform useful functions and are safe to come in contact with since, 64
- Nonporous**, an item that is made or constructed of a material that has no pores or openings and cannot absorb liquids, 60

Normal skin, 320
 Nose muscles, 154
 Nose shapes, 390
 No-stem, 542
 Notching, 411, 538
 Note taking, 34
 Nubian coils, 483–485
Nucleus, dense, active protoplasm found in the center of a cell; important to reproduction and metabolism, 145
 Nutrition, 41–42

O

Objections, overcoming, 748–749
 Oblong-shaped face, 389
Occipital artery, artery that supplies the scalp and back of the head up to the crown, 164, 309
Occipital bone, hindmost bone of the skull; located below the parietal bones, 149
Occipitalis, back of the epicranium; muscle that draws the scalp backward, 152
Occupational disease, illness resulting from conditions associated with employment, such as prolonged and repeated overexposure to certain products or ingredients, 68
 Occupational Safety and Health Act of 1970, 58
 Occupational Safety and Health Administration (OSHA), 57–58, 759
Off-base, position of a curl off its base; provides maximum mobility and minimum volume, 542
 Offset handle design, 111
Off-the-scalp lighteners, lighteners that cannot be used directly on the scalp, 658
Ohm (O), the unit of electrical resistance to an electric current, 208
 Oil glands, 228
 Oil lighteners, 658, 670
 Oil-base dye removers, 660
 Oily scalp and hair, treatment for, 284, 298
 Oily skin, 320–321, 325
 Ointments, 198
On-base, position of a curl directly on its base; provides maximum volume, 542
Once-over shave, single-lather shave in which the shaving strokes are made across the grain of the hair, 349
 180-degree elevation, 399
 Online portfolios, 721
Onyx, technical term for nail, 782
On-the-scalp lighteners, lighteners that can be used directly on the scalp, 657
Open comedo, also known as a *blackhead*; a hair follicle filled with keratin and sebum, 234
Ophthalmic nerve, supplies impulses to the skin of the forehead, upper eyelids, and interior portion of the scalp, orbit, eyeball, and nasal passage, 159

Opposing handle design, 111
Orbicularis oculi muscle, ring muscle of the eye socket, 154
Orbicularis oris muscle, flat band around the upper and lower lips, 154
Organic chemistry, the study of substances that contain carbon, 179
Organization, a method used to store new information for short- and long-term memory, 32
 Organizational plan, 763
Organs, structures composed of specialized tissues performing specific functions, 147
Origin, fixed part of a muscle that does not move, 151, 152
Os, means “bone”; it is used as a prefix in many medical terms, 148
Osteology, the study of the anatomy, structure, and function of the bones, 148
 Outline shave, 349, 416, 428–430
 Outliners, 89–92, 114–117
Outlining, finish work of a haircut with shears, trimmers, or razor, 401
 Oval-shaped face, 388
Overdirection, combing a section away from its natural falling position, rather than straight out from the head, toward a guideline; used to create increasing lengths in the interior or perimeter, 402, 537
 Overprocessing, 600
 Ownership, types of, 760–762
Oxidation, either the addition of oxygen or the loss of hydrogen; a chemical reaction that combines an element or compound with oxygen to produce an oxide, 182, 183
 Oxidation haircolor, 182
 Oxidation tints, 653–654
Oxides, compounds of any element combined with oxygen, 184
 Oxidizing agents, 182
 Oxy melanin, 655

P

Packs, 198
 Paddle brush, 108
 Pain receptor, 226
Palming the comb, the technique used to hold the comb in the hand opposite of the hand that is cutting with the shears, 113
Palming the shears, the technique used to hold shears in a safe manner while combing through or otherwise working with hair, 113
Papillary layer, outer layer of the dermis, directly beneath the epidermis, 225
Papule, a small elevation on the skin that contains no fluid, but may develop pus, 231

Para-aminobenzoic acid (PABA), 199
 Paraffin wax masks, 323
Parasites, organisms that grow, feed, and shelter on or inside another organism (referred to as the *host*), while contributing nothing to the survival of that organism; parasites must have a host to survive, 64, 71
Parasitic disease, disease caused by parasites, such as lice and mites, 68
 Parasitic infestations, 263
 Paré, Ambroise, 12
Parietal artery, artery that supplies blood to the side and crown of the head, 164
Parietal bones, bones that form the sides and top of the cranium, 149
Parietal ridge, widest area of the head, also known as the crest, hatband, horse-shoe, or temporal region, 393
Part, a line, created naturally or with a comb, that divides the hair at the scalp, separating one section of the hair from another, 399
Parting, a line dividing the hair of the scalp that separates one section of the hair from another or creates subsections from a larger section of hair, 399–400
Partnership, business structure in which two or more people share ownership, although not necessarily equally, 760–761
 Pastes, 198
Patch test, test for identifying a possible allergy to aniline derivative products; required by the FDA 24 to 48 hours before the application of a product, 662, 678–679
Pathogenic, harmful microorganisms that can cause disease or infection in humans when they invade the body, 64
 Pathogenic bacteria, 65
Pathogenic disease, disease produced by organisms, including bacteria, viruses, fungi, and parasites, 68
 Pathogens, 64
 Pear-shaped face, 388–389
Pectoralis major and pectoralis minor, muscles of the chest that assist the swinging movements of the arm, 155
Pediculosis capitis, infestation of the hair and scalp with head lice, 71, 263
Peptide bonds, end bonds; chemical bonds that join amino acids end to end, 251
Percussion, another name for tapotement, 311–312
Perfectionism, an unhealthy compulsion to do things perfectly, 26
 Periodic table, 179
Perionychium, the living skin bordering the root and sides of a nail, 783

Peripheral nervous system (PNS), made up of sensory and motor nerve fibers that extend from the brain and spinal cord to all parts of the body. Their function is to carry impulses or messages to and from the CNS, 157

Permanent haircolor, lighten and deposit color at the same time and in a single process because they are more alkaline than no-lift, deposit-only colors and are usually mixed with a higher-volume developer, 649, 652–654, 669

Permanent wave, 253, 506–507, 524

Permanent wave neutralizer, 182

Permanent wave solution, 182

Permanent waving, a process used to chemically restructure natural hair into a different wave pattern, 578–579

Personal budget, 744–745

Personal grooming, the process of caring for parts of the body and maintaining an overall polished look, 42–44

Personal hygiene, daily maintenance of cleanliness by practicing good healthful habits, 41

Personnel, your staff or employees, 770–772

Pétrissage, kneading movement performed by lifting, squeezing, and pressing the tissue with a light, firm pressure, 311

Petrolatum, 200

Petroleum distillates, 76

pH, (potential hydrogen), relative degree of acidity or alkalinity of a substance, 187–190

pH scale, a measure of the concentration of hydrogen ions in acidic and alkaline solutions, 188

Phalanges, bone of the fingers or toes, 151

pH-balanced shampoos, shampoos that are balanced to the pH of hair and skin (4.5 to 5.5); also known as acid-balanced shampoos, 193

Phenolic disinfectants, tuberculocidal disinfectants that are a form of formaldehyde, have a very high pH, and can damage the skin and eyes, 60, 75

Pheomelanin, melanin that gives red to blond colors to hair, 253, 644

Phototherapy, 216

Physical change, change in the form of a substance without the formation of a new substance, 181

Physical mixture, combination of two or more substances united physically, 183–184

Physical presentation, your posture, as well as the way you walk and move, 44–47, 280

Physical properties, characteristics of matter that can be determined without a chemical reaction, 181

Physiology, study of the functions or activities performed by the body's structures, 144

body systems and associated organs, 147–167

cells, 145–146

circulatory system, 161–165

endocrine system, 166–167

integumentary system, 165–166

lymphatic/immune system, 165

muscular system, 151–157

nervous system, 157–160

reproductive system, 167

tissues, types of, 147

Pick combs, 106

Pictograms, 59

Pigment disorders, 237–238

Pin curls, 540

Pityriasis, technical term for dandruff, 261

Pityriasis capitis simplex, dry dandruff, 262

Pityriasis steatoides, waxy or greasy dandruff, 262

Pivot motor clippers, 115–116

Pivot screw, 109

Plaster mold form, 504, 511–513

Platysma, broad muscle that extends from the chest and shoulder to the side of the chin; depresses lower jaw and lip, 153

PNS. See Peripheral nervous system (PNS)

Point cutting, 411, 538

Polarity, negative or positive pole of an electric current, 211, 212

Polypeptide chain, long chain of amino acids linked by peptide bonds, 251

Pomades, 674

Pompadour, 414

Pompadour fade, 459–464

Pope Alexander III, 11

Porosity, 267–268

Porous, made or constructed of a material that has pores or openings; porous items are absorbent, 78

Port wine stain, 238

Positive approach, 48

Positive personality and attitude, 36

Positive space, 397

Posterior auricular artery, artery that supplies blood to the scalp, behind and above the ear, 164, 309

Posterior auricular nerve, nerve that affects the muscles behind the ear at the base of the skull, 160

Posture, 44–45

Potassium and lithium hydroxide relaxers, 606

Potential hydrogen, 188

Powder lighteners, 658, 670

Practical exams, hands-on testing on a live model or mannequin, 713–714

Precision cut, 414, 431–438

Pre-custom systems, 501

Predisposition test, 662

Pre-lightening, the first step of a double-process haircoloring; used to lighten natural pigment, 669

Preservative treatments, 320

Pre-softening, process of treating resistant hair for better color penetration, 663

Press-button can latherizers, 129

Pressing combs, 130, 131

Pressure receptor, 226

Pre-wrap solution, usually a type of leave-in conditioner that may be applied to the hair prior to permanent waving to equalize porosity, 598

Primary colors, red, blue, and yellow; colors that cannot be achieved from a mixture of other colors, 645

Primary skin lesions, lesions that are a different color than the color of the skin or lesions that are raised above the surface of the skin, 230–232

Print ads, 497

Prioritize, to make a list of tasks that need to be done in the order of most-to-least important, 30

Problem solving, 736

Procerus, muscle that covers the bridge of the nose, depresses the eyebrows, and wrinkles the nose, 154

Procrastination, putting off until tomorrow what you can do today, 26

Productivity, 26

Professional advice, 745–746

Professional ethics, 35–36

Professional image, the impression you project through both your outward appearance and your conduct in the workplace

definition of, 42

ergonomics, 44–47

healthful habits, 40–42

human relations and communication skills, 47–50

image-building basics, 42–44

social media, 50

Professionalism, 750

Profiles, 390

Progressive colors, haircolor products that contain compound or metallic dyes, which build up on the hair; not used professionally, 654

Projection, angle or elevation that hair is held at from the head for cutting, 398–402

Prominent nose shapes, 390

Pronator, muscle that turns the hand inward so that the palm faces downward, 156

Proportion, a design principle that measures or shows the comparative relationship between two or more design elements of a form, 396

Proteins, 251

Protoplasm, 66

Protoplast, a colorless jellylike substance found inside cells in which food elements such as proteins, fats, carbohydrates, mineral salts, and water are present, 145

Pseudofolliculitis barbae, a chronic inflammatory form of folliculitis known as razor bumps resembling folliculitis papules and pustules; generally accepted to be caused by ingrown hair, 264

Psoriasis, skin disease characterized by red patches and silvery-white scales, 236

Psychology of success, 25–27

Pterygoid muscles, 153

Public speaking, 751

Pulmonary circulation, takes deoxygenated blood to the lungs for oxygenation and waste removal and then returns that blood to the heart, so oxygen-rich blood can be delivered to the body, 161

Punctuality, 736

Pure substance, matter that has a fixed chemical composition, definite proportions, and distinct properties, 183–184

Pus, a fluid created by infection, 67

Pustule, raised, inflamed, papule with a white or yellow center containing pus in the top of the lesion, 231

Pythagoras, 9

Q

Quadratus labii inferioris muscle, 154

Quadratus labii superioris muscle, 154

Quaternary ammonium compounds, commonly known as quats are products made of quaternary ammonium cations and are designed for disinfection of nonporous surfaces; they are appropriate for use in noncritical (noninvasive) environments and are effective against most pathogens of concern in the barber shop environment, 60, 75, 200

Quaternary colors, 645

Quo vadis, 415

R

Radio waves, 213–214

Radius, smaller bone in the forearm on the same side as the thumb, 151

Rapid oxidation, 183

Razor balance, 121

Razor bumps, 264

Razor care, 122

Razor cut, 444–448

Razor cutting

hair thinning and texturizing, 410
removing weight from the ends, 410–411
strokes, 408–410
techniques, 408
women's haircutting, 537–538

Razor finish, 122

Razor grind, 121–122

Razor quality, 121

Razor rotation, texturizing technique similar to razor-over-comb, done with small circular motions, 408

Razor shapers, 120

Razor temper, 121

Razor-over-comb, texturizing technique in which the comb and the razor are used on the surface of the hair, 408

Razors

history of, 10
honing, 133–134
position and stroke practice, 355–357
positions and strokes, 342–344
size, 122
stropping, 125–126, 135
style, 122

Rebooking clients, 751–752

Receptivity, 36

Receptors, 158

Reclined method, 279

Reconditioning damaged hair, 673–674

Record keeping, 664–665

Record keeping, maintaining accurate and complete records of all financial activities in your business, 760, 765–767

Rectifier, apparatus that changes AC to DC, 207

Redox, term used for an oxidation-reduction reaction, 182–183

Reduction, the subtraction of oxygen from, or the addition of hydrogen to, a substance, 182, 183

Reference points, points on the head that mark where the surface of the head changes or the behavior of the hair changes, such as ears, jawline, occipital bone, and apex; used to establish design lines that are proportionate, 393

Referrals, 496

Reflex, automatic nerve reaction to a stimulus, 158

Relationship building, 737

Relaxer strengths, 607

Reliability, 750

Repetition, repeatedly saying, writing, or otherwise reviewing new information until it is learned, 32, 49

Reporting income, 743–746

Reproductive system, body system responsible for processes by which plants and animals produce offspring, 148, 167

Respect, 26

Respiratory system, consists of the lungs and air passages; enables breathing, 148

Rest and relaxation, 41

Resting phase, 256

Resume, written summary of a person's education and work experience, 716–719, 720

Retail supplies, supplies sold to clients, 766

Retailing, the act of recommending and selling products to your clients for at-home use, 746

Reticular layer, deeper layer of the dermis, 225

Retouch application, application of the product to new growth only, 653, 662

Reverse backhand, razor position and stroke used by right-handed barbers for shaving the left side of the neck behind the ear and used by left-handed barbers behind the right ear, 342–343, 346–347, 357

Reverse freehand, razor position and stroke used in 4 of the 14 basic shaving areas: nos. 5, 10, 13, and 14, 342–343, 345–346, 357

Reverse highlighting, 663

Rheostat, an adjustable resistor used for controlling current in a circuit, 208

Rhymes, 32

Ridges, depressions running vertically down the length of the nail, 389, 600, 782

Right-handed barbers, shaving movements for, 341–342

Right-handed shears, 113

Rinse-out conditioners, 194

Risorius muscle, muscle of the mouth that draws the corner of the mouth out and back as in grinning, 154

Rogaine, 260, 498

Role models, 742

Roller sets, 540

Rolling cream, cleansing and exfoliating product used in facials to lift dead skin cells and dirt from the skin surface, 322

Rolling the comb out, a method used to put the hair into position for cutting by combing into the hair with the teeth of the comb in an upward direction, 404

Roman Catholic Church, 9

Root-turning, refers to sorting the hair strands so that the cuticle points toward the hair ends in its natural direction of growth, 501

Rosacea, chronic congestion of the skin characterized by redness, blood vessel dilation, papules, and pustules, 235

Rotary (universal) motor clippers, 115–116

Rotary movement, 282

Round brush, 108
 Round-shaped face, 388
 Rules, vs. laws, 62

Russian strop, a type of cowhide strop that is considered to be one of the best and that requires breaking in, 125

S

S Corporation, 761

Safe work practices, 80–84

Safety Data Sheet (SDS), required by law for all products sold. SDSs include safety information about products compiled by the manufacturer, including hazardous ingredients, safe use and handling procedures, proper disposal guidelines, and precautions to reduce the risk of accidental harm or overexposure, 58–60, 76, 84, 759

Safety devices, 209–211

Safety precautions, 80–84

Safety with electrical equipment, 209–211

Salary-plus-commission, 740

Salt bond, a physical side bond within the hair cortex, 251, 252

Salts, compounds that are formed by the reaction of acids and bases, 184

Sanitation, also known as *sanitizing*, a chemical process for reducing the number of disease-causing germs on cleaned surfaces to a safe level, 56

Saturation, 647

Scabies, a contagious skin disease that is caused by the itch mite, which burrows under the skin, 71, 263

Scale, an accumulation of dry or greasy flakes on the skin, 233

Scalp. *See* Hair and scalp disorders; Hair and scalp treatment

Scalp analysis, 265–266, 280, 580–581

Scalp conditioners, cream-based products and ointments used to soften and improve the health of the scalp, 196, 277

Scalp lotions, 199

Scalp massages, 257, 282–283, 295–296

Scalp muscles, 152–155

Scalp reduction, the surgical removal of a bald area, followed by the pulling together of the scalp ends, 499

Scalp steam, process of using steam towels or a steaming unit to soften and open scalp pores, 284–285, 294

Scalp treatment variations, 297–300

Scar, also known as *cicatrix*, slightly raised or depressed mark on the skin formed after an injury or a lesion of the skin has healed, 233

School release form, 665

Scientific rest facial, 324

Scrunch styling, 417–418

Scutula, dry, sulfur-yellow, cuplike crusts on the scalp seen in *tinea favosa*, 263

Sebaceous cyst, a large, protruding pocket-like lesion filled with sebum, frequently seen on the scalp and back, 234

Sebaceous glands, oil glands of the skin connected to hair follicles, 228, 234–237, 249, 325

Seborrheic dermatitis, is a skin condition caused by an inflammation of the sebaceous glands, and is often characterized by redness, dry or oily scaling, crusting, and/or itchiness, 235, 262

Sebum, an oily substance secreted by the sebaceous glands, 228, 235, 249

Secondary colors, colors obtained by mixing equal parts of two primary colors, 645

Secondary skin lesions, lesions characterized by an accumulation of material on the skin surface, such as a crust or scab, or by depressions in the skin surface, 230, 232–233

Second-time-over shave, follows a regular shave to remove any rough or uneven spots using water instead of lather; may be considered a form of close shaving, 348–349

Secretions, 229

Secretory nerve fibers, nerves regulate the excretion of perspiration from the sweat glands and the flow of sebum from the oil glands, 225

Sectioning, 592–593

Selective photothermolysis, 216

Self-actualization, fulfilling one's full potential, 36

Self-care, 35

Self-care test, 35

Self-esteem, 25

Selling, 746–749, 778

overcoming objections, 748–749

principles of, 747–748

psychology of, 748–749

Semi-convex blade, 110

Semipermanent haircolor, deposit-only haircolor product formulated to last through several shampoos, 649, 651, 667–668

Sensation, 229

Sensitivity, 36

Sensory nerve fibers, nerves that react to heat, cold, touch, pressure, and pain and send messages to the brain, 225, 226

Sensory nerves, nerves that carry impulses or messages from the sense organs to the brain, 158

Serratus anterior, muscle of the chest that assists in breathing and in raising the arm, 155

Service profession, thriving in, 736

Set of the shears, the manner in which the blades and shanks of the shears align

with each other and are joined at the tension screw or rivet, 110

Set screw, 115

Seventh cranial nerve, chief motor nerve of the face, 159–160, 308

Shadow fade cut, 469–472

Shampoo, hair and scalp cleansing products, 277

application of, 281

balancing, 193

chemistry, 190–191

clarifying, 193

classification of, 191–192

color-enhancing, 193

conditioning, 193

dry or powder shampoo application, 293

for hair replacements and wigs, 194

liquid-dry shampoo application, 293

matching to hair types, 196

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neutralizing shampoos, 194

pH-balanced, 193

sulfate-free, 194

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Shampoo capes, 127, 278

Shampoo service, 276–277

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draping methods, 278

shampoo and shampoo massage

manipulations, 290–292

shampooing and rinsing methods, 279

Shank, 109, 110

Shave service, 278, 348–349

Shaving, 338–339

backhand position and stroke, 345

basic guidelines, 339–340

beard design, 375–380

body positioning, 347

close shaving, 349

customer satisfaction, 340

first-time-over shave, 348

freehand position and stroke, 344–345

hair growth considerations, 190

hair type considerations, 339

handling a straight razor, 353–354

infection control and safety precautions, 352

movements for left-handed and right-handed

barbers, 341–342

mustache trim, 374

neck shave, 349, 372–373

once-over shave, 349

razor position and stroke practice, 355–357

razor positions and strokes, 342–344

reverse-backhand position and stroke,

346–347

reverse-freehand position and stroke, 345–346

second-time-over shave, 349

shaving areas of the face, 340–342

steps in, 348, 358–371

Shaving soaps, 129

Shear blades, 109

Shear cutting techniques, 403–405

Shear-over-comb, haircutting technique in which the hair is held in place with the comb while the shears are used to remove length, 403–404

Shear-over-comb position, 107

Shear-point tapering, haircutting technique used to thin out difficult areas in the haircut, such as dips and hollows, 404

Shears, 112–114. *See also* Blending shears;

Haircutting shears; Texturizing shears;

Thinning shears

Shell strop, a type of horsehide strop made from the muscular rump area of the horse and considered to be the best strop for use by barbers, 125

Short-term goals, 29

Shoulder bones, 150–151

Shoulder muscles, 155–156

Side bonds, also known as cross bonds; hydrogen, salt, and sulfur bonds in the hair cortex, 251–253

Sideburn lengths and designs, 391–392

Silicones, 195, 200

Single flat wrap mini-procedure, 591

Single-process haircoloring, process that lightens and colors the hair in a single application, 662

Single-process permanent color, 686–688

Single-process tints, 669

Skeletal system, physical foundation of the body; composed of bones and movable and immovable joints, 148–151

Skin

absorption level, 228

acne treatment, 235

analysis of, 321

anatomy of, 222–229

color of, 226

common inflammations and infections, 236–237

dermis, 224–225

disorders and diseases, 229–233

disorders of the sebaceous and sudoriferous glands, 234–237

elasticity of, 226

epidermis, 223–224

fluids of, 225

functions of, 228–229

glands of, 227–228

hypertrophies of, 238–239

lesions of, 230

maintaining health of, 240

nerves of, 225–226

pigment disorders, 237–238

types of, 320–324

Skin cancer, 237, 239–240

Skin care, 43

Skin care products, 322–324

Skin elasticity, 226

Skin freshening lotions, 197

Skin stain removers, 661

Skin structures, 166

Skin tag, a small brown or flesh-colored outgrowth of the skin occurring most frequently on the neck and chest, 239

Skin tonics, toners, fresheners, and astringents; products used to help rebalance skin pH, remove product residue, and create a temporary tightening effect on the skin, 197, 322

Skin types, 320–324

Skull, 149

Sleep, 41

Slicing, 539

Slicing and carving, 410

Sliding movement, 282

Slithering, 410, 538–539

Small to average nose shapes, 390

Smaller occipital nerve, nerve that affects the scalp and muscles behind the ear, 160

Soap cap, equal parts of tint and a shampoo, 662

Soaps, 199

Social media, a platform used to engage and communicate with groups of people through online communities, networks, websites, or blogs, for personal or professional means, 50, 496, 751, 775–776

Sodium hydroxide relaxers, 606

Sodium hypochlorite, common household bleach; an effective disinfectant for the barbershop, 75

Soft press, 542

Soft water, 187

Sole proprietor, individual owner and, most often, the manager of a business, 760

Solubility, 185

Solute, any substance that is dissolved into a solvent to form a solution, 185

Solution, a stable, uniform mixture of two or more mixable substances made by dissolving a solid, liquid, or gaseous substance in another substance, 185, 186

Solvent, any substance, usually a liquid, that dissolves the solute to form a solution, 185

Songs, 32

Space, 397

Special needs, shampooing clients with, 279–281

Special-effects haircoloring, 672, 694–698

Specialized ligaments, bands of fibrous tissue that attach the nail bed and matrix to the underlying bone that are located at the base of the matrix and around the edges of the nail bed, 783

Sphenoid bone, bone that connects all the bones of the cranium, 149

Spinal cord, portion of the central nervous system that originates in the brain and runs downward through the spinal column, 157

Spiny layer, 224

Spiral wrap, 594

Spirilla, spiral or corkscrew-shaped bacteria that cause diseases such as syphilis and Lyme, 65

Split ends, 261

Spore test, 72

Spore-forming, the ability of some bacteria to form a protective coating around themselves to protect them from harsh environments during the inactive stage, 66

Squamous cell carcinoma, type of skin cancer more serious than basal cell carcinoma, but not as serious as malignant melanoma, 239

Square-shaped face, 388

Stain, abnormal brown- or wine-colored skin discoloration, 238

Standard Precautions, are guidelines published by the CDC that require the employer and employee to assume that any human blood and body fluids are potentially infectious, 80

Staphylococci, pus-forming bacteria that grow in clusters like a bunch of grapes; they cause abscesses, pustules, and boils, 65, 66, 67, 264

Star design, fade with, 465–468

State regulatory agencies, 61

Stationary guide, guideline that does not move, but all other hair is brought to it for cutting, 400

Steatoma, sebaceous cyst or fatty tumor, 235

Stem, the section of a curl between the base and the first arc of the circle; gives the curl direction and movement, 541; the basic question or problem, 711

Sterilization, the process that completely destroys all microbial life, including spores, 72

Sternocleidomastoideus, muscle of the neck that depresses and rotates the head, 153

Still blade, 109, 115

Still point, 109

Stock systems, 501

Straight hair, 254

Straight lines, 395

Straight profiles, 390

Straight razors, 119–120

conventional straight razor, 121–122

haircutting, 120

handling, 353–354

holding, 120–121

honing and stropping, 120

shaving, 120

Straight-blade trimmer, 116

Strand test, the application of a coloring or lightening product to determine how the hair will react to the formula and the amount of time it will take to process, 607–608, 662, 680–681

Stratum corneum, outermost layer of the epidermis; the horny layer, 223, 224

Stratum germinativum, innermost layer of the epidermis, also known as the *basal* or *Malpighian layer*, 224

Stratum granulosum, granular layer of the epidermis beneath the stratum lucidum; the grainy layer, 224

Stratum lucidum, clear layer of the epidermis, directly beneath the stratum corneum, 223, 224

Stratum spinosum, spiny layer of the epidermis, often considered part of the stratum germinativum, 224

Streptococci, pus-forming bacteria arranged in curved lines resembling a string of beads. They cause infections such as strep throat and blood poisoning, 65

Stress management, 42

Strop, an elongated piece of leather or other materials used to finish the edge of conventional straight razors to a smooth, whetted cutting edge, 78, 123, 124–126, 135

Strop dressing, 125

Styling aids, 196–197, 199

Styling chairs, 126

Styling techniques, 417–421

Stylized blowdrying, 418–419, 478

Styptic powder, alum powder or liquid used to stop bleeding of nicks and cuts, 200, 352

Subcutaneous tissue, fatty tissue layer that lies beneath the dermis; also called *adipose tissue*, 225

Subdermal systems, 304–309

- modalities that affect muscle action, 305
- modalities that affect nerve responses, 308
- muscles affected by facial massage, 305–308

Submental artery, artery that supplies blood to the chin and lower lip, 164

Success, 25–27

Sudoriferous glands, sweat glands of the skin, 227, 228, 234–237

Sulfate-free shampoos, formulated with little to no sulfates so does not strip natural oils from the hair, 194

Sulfur bond, 252

Sun protection factor (SPF), 199

Suntan lotions, 199

Superficial temporal artery, artery that supplies blood to the muscles of the front, sides, and top of the head, 164, 309

Superior labial artery, artery that supplies blood to the upper lip and lower region of the nose, 164

Supinator, muscle of the forearm that rotates the radius outward and the palm upward, 156

Supplements, 240

Supraorbital artery, artery that supplies blood to the upper eyelid and forehead, 164

Supraorbital nerve, nerve that affects the skin of the forehead, scalp, eyebrows, and upper eyelids, 159

Supratrochlear nerve, nerve that affects the skin between the eyes and the upper side of the nose, 159

Surface active agents, 185

Surfactants, a substance that acts as a bridge to allow oil and water to mix or emulsify by reducing surface tension, 186–187

Surgical hair restoration, 498–499

Suspensions, formulations in which solid particles are distributed throughout a liquid medium, 185, 186

Swayback, 46

Sweat glands, 227

Sycosis vulgaris, chronic bacterial infection of the bearded areas of the face, 264–265

Synthetic bristle brush, 108

Synthetic elements, 179

Synthetic hair, 499–500

Synthetic bones, 123

Systemic circulation, carries the oxygen-rich blood from the heart throughout the body and returns deoxygenated blood back to the heart, 161

Systemic disease, 68

T

Tactile corpuscles, 225

Tail combs, 106

Tan, darkening of the skin due to exposure to ultraviolet rays, 238

Tangled cord syndrome, 82

Taper, haircuts in which there is an even blend from very short at the hairline to longer lengths as you move up the head; *to taper* is to narrow progressively at one end, 393–394, 401

Taper combs, used for cutting or trimming hair when a gradual blending from short to longer is required within the haircut, 106

Taper cut, 413

Taping up, 386

Tapotement, most stimulating massage movement, consisting of short, quick tapping, slapping, and hacking movements, 311–312

Teamwork, 736–737

Telangiectasis, distended or dilated surface blood vessels, 235

Telephone, use of, 774–775

Telogen phase, resting phase of the hair growth cycle, 256

Temple fade, 455–458

Temporal bones, bones that form the sides of the head in the ear region, 149

Temporal nerve, nerve that affects the muscles of the temple, side of the forehead, eyelid, eyebrow, and upper cheek, 160

Temporalis, muscle that aids in opening and closing the mouth and chewing, 153

Temporary color rinse, 667, 682–683

Temporary colors, color products that last only from shampoo to shampoo, 649, 650–651

Tension, amount of pressure applied when combing and holding a section, created by stretching or pulling the section, 401

Tension screw, 110

Terminal blending, 409

Terminal hair, long hair found on the scalp, beard, chest, back, and legs, 255

Tertiary colors, 645

Tesla, Nikola, 131

Tesla high-frequency current, thermal or heat-producing current with a high oscillation rate; also known as the *violet ray*, 213, 315

Test curls, 601, 612–613

Test-wise, understanding the strategies for successful test taking, 709

Texture, 267, 535

Texturize, a process used to semi-straighten extremely curly hair into a more manageable texture and wave pattern, 610–611

Texturizing, removing excess bulk without shortening the length; changing the appearance or behavior of hair through specific haircutting techniques using shears, thinning shears, clippers, or a razor, 401, 538–539

Texturizing shears, shears with 14 to 28 medium width teeth that produce visible differences in hair lengths to add texture and volume, 111–112

Thank-you note, 724

Therapeutic lamps, 217

Therapeutic medicated shampoos, contain special chemicals or drugs that are very effective in reducing excessive dandruff, 194

Thermal irons, 130–131, 138–139

Thermal straightening, 542–543

Thermal styling, methods of curling or straightening dry hair using thermal irons and/or pressing combs, 541–543

Thermal styling tools, refers to tools that produce heat for hairstyling purposes, 105, 128, 130.

Thermal waving, 541–542

Theroxidil, 260

Thinning, removing bulk from the hair, 401, 410

Thinning shears, also known as *blending shears*; shears with 30 to 50 thin, narrowly spaced teeth that eliminate visible lines in the hair, used for blending hair ends and removing bulk or weight, 111–112

Thio relaxers, relaxers that usually have a pH above 10 and a higher concentration of ATG than is used in permanent waving, 587, 605–608, 610

Thio-free waves, 598

Thorax, the chest, 150

Thumb grip, 109, 110

Ticinius Mena, Sicilian credited with bringing barbering and shaving to Rome in 296 BC, 8

Ticket upgrading, Also known as *upselling services*; the practice of recommending and selling additional services to your clients, 746

Time management, 30–31

Tinea, technical name for ringworm, 262–263

Tinea barbae, also known as barber's itch, a superficial fungal infection that commonly affects the skin; it is primarily limited to the bearded areas of the face and neck or around the scalp, 70, 263

Tinea capitis, a fungal infection of the scalp characterized by red papules, or spots, at the opening of the hair follicles; 70, 262

Tinea favosa, ringworm characterized by dry, sulfur-yellow crusts on the scalp, 263

Tinea sycosis, ringworm of the bearded areas on the face, 265

Tint, 660

Tint back, 662

Tint or dye solvents, 660

Tinted hair, 603

Tips, 740

Tissue, collections of similar cells that perform a particular function, 147

Tone, term used to describe the warmth or coolness of a color, 646, 647–648

Toners, skin tonics with an alcohol content of 4 to 15 percent; most are designed for use on normal and combination skin types, 322; semipermanent, demipermanent, or permanent hair-color products used primarily on pre-lightened hair to achieve pale and delicate colors, 653, 659–660, 671

Tonsorial, related to the cutting, clipping, or trimming of hair with shears or a razor, 8

Tonsure, a shaved patch on the head, 9

Tools, 77, 81–82, 86–88

Topical medications for hair replacement, 498

Touch receptor, 226

Toupee, outdated term used to describe a small hair replacement that covers the top or crown of the head, 494

Towel wrapping, 136–137

Towels, 77–78, 127–128

Toxins, various poisonous substances produced by some microorganisms (bacteria and viruses), 64, 71

Transferable skills, skills mastered at other jobs that can be put to use in a new position, 717, 718

Transverse facial artery, artery that supplies blood to the skin and the masseter, 164

Trapezius, muscle that covers the back of the neck and upper and middle region of the back, 155

Traveling guide, guideline that moves as the haircutting progresses; used when creating layers or graduation; also referred to as moving or movable guidelines, 400

Treatment or repair conditioners, deep, penetrating formulations that help restore moisture and protein in the hair, 195

Triangularis muscle, muscle that extends alongside the chin and pulls down the corner of the mouth, 154

Tricep, large muscle that covers the entire back of the upper arm and extends the forearm, 156

Trichology, the science dealing with the hair, its diseases, and its care, 248

Trichoptilosis, the technical term for split ends; hair conditioning and oils will smooth split ends, but the only way to remove them is to cut them, 261

Trichorrhexis nodosa, technical term for knotted hair, 261

Trifacial nerve, 158, 308

Trigeminal nerve, 158, 308

Trigger point, a tender area in a muscle caused by a localized knot or spasm in the muscle fiber that can radiate pain to other locations in the body, 311

Trimmers, small clippers, also known as outliners and edgers, used for detail, precision design, and fine finish work after a haircut or beard trim, 114–117

Trims, 386–387

True acid waves, perms that have a pH between 4.5 and 7.0 and require heat to speed processing; process more slowly than alkaline waves and do not usually produce as firm a curl, 596–597

True skin, 224

True/false questions, 711

T-shaped blade trimmer, 116

Tubercle, abnormal solid lump above, within, or below the skin, 230

Tuberculocidal disinfectants, often referred to as phenolics, are proven to kill the bacterium that cause tuberculosis, in addition to other pathogens destroyed

through the use of hospital disinfectants, 60, 75

Tuberculosis, a disease caused by bacteria that are transmitted through coughing or sneezing, 60

Tumor, abnormal cell mass resulting from excessive multiplication of cells, 231

Tweezer, 132

U

Ulcer, open skin lesion accompanied by pus and loss of skin depth; a deep erosion; a depression in the skin, normally due to infection or cancer, 233

Ulna, inner and larger bone of the forearm, 151

Ultraviolet-ray lamps, 318

Ultraviolet light, invisible rays, with short wavelengths and minimal skin penetration, that produce chemical effects and kill germs; also called *actinic* or *cold rays*, 215, 217

Underlying pigment, 658

Underprocessing, 600

Undertone, 644

Uniform-layered cut, haircut in which all the hair is cut at the same length using a 90-degree elevation, 534, 554–558

Unisex salons, 17

United States Pharmacopeia (USP), 199–200

Universal Precautions (UP), 80

Universal solvent, 187

Upselling services, 746

V

Values, 36

Vascular system, 161–165

Vegetable tints, 654

Veins, thin-walled blood vessels that are less elastic than arteries; veins contain cuplike valves that keep blood flowing in one direction to the heart and prevent blood from flowing backward. Veins carry blood containing waste products back to the heart and lungs for cleaning and to pick up oxygen, 161, 305

Vellus hair, soft, downy hair that appears on the body, 255

Vented brush, 108

Ventilation, 82

Venules, small vessels that connect the capillaries to the veins; they collect blood from the capillaries and drain it into the veins, 161

Verruca, technical name for a wart, 239

Vertical lines, lines that are straight up and down, 395

Vesicle, small blister or sac containing clear fluid, 232

Vibration, in massage, the rapid shaking of the body part while the fingertips are pressed firmly on the point of application, 312

Vibratory facial, 324

Violet ray, 213

Virgin application, the first time the hair is tinted or lightened, 662

Virucidal, capable of destroying viruses, 63

Virus, a parasitic submicroscopic particle that infects and resides in cells of biological organisms; a virus is capable of replication only through taking over the host cell's reproductive function, 64, 67–69

Visible light, electromagnetic radiation that can be seen by the human eye, 215

Visible spectrum of light, electromagnetic spectrum that can be seen by the human eye, 215

Vision statement, a long-term picture of what the business is to become and what it will look like when it gets there, 757, 763

Visual study skills, 33–34

Vitamins, 240

Vitiligo, an acquired leukoderma characterized by milkywhite spots, 238

Vocabulary, 27

Volt (V), a unit of electrical pressure that pushes the flow of electrons forward through a conductor, 208

Volume, building, 420

Volume, the measure of the potential oxidation of varying strengths of hydrogen peroxide, 655

W

Wage structures, 739–741

Warm colors, 646

Wart, also known as *verruca*; an infectious hypertrophy of the papillae and epidermis caused by a virus, 239

Water, 81, 187–190

Water temperature, 281

Water wrap, 594

Waterless hand sanitizers, 79

Watt (W), the unit of power (amperes multiplied by volts), indicating how much electric energy is being used in 1 second, 209

Wave formation and processing, 600–601

Wave pattern, amount of movement in the hair strand; straight, wavy, curly, and coiled, 253–254

Wave rods, 588–589

Wave wrap, 588–595

Wavelength, distance between two successive peaks of electromagnetic waves, 214

Wavy hair, 254

Wedge grind, 121

Weight line, a visual line in the haircut, where the ends of the hair hang together; the line of maximum length within the weight area; heaviest perimeter area of a 0-degree (one-length) or 45-degree (graduated) cut, 401

Wen, 235

Wet brush, 108

Wet capes, 278

Wet hairstyling, 540

Wet setting, 252

Wheal, itchy, swollen *lesion* caused by insect bites or plant irritations, such as nettle, 232

Whitehead, a closed comedone; consists of accumulated sebum that remains a whitish color because it does not have a follicular opening for exposure to oxygen, 234

Whorl, hair that grows in a circular pattern, 257

Wide-tooth combs, 106

Wig block, also known as *styling*; a head-shaped form made of plastic, foam, or other materials used as a stand for a wig or hair replacement system, 503, 521–523

Wigs, 523

Window displays, 496–497

Wiry hair, 267

Witch hazel, 200

Women's haircutting and styling

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- vs. men's, 531

techniques for cutting natural curly styles, 536–537

thermal styling, 541–543

uniform-layered cut, 534, 554–558

wet hairstyling, 540

Word associations, 32

Word of mouth, 496

Work ethic, taking pride in your work and committing yourself to consistently doing a good job for your clients, employer, and barbershop team, 716

Work surfaces, disinfecting, 77

Worshipful Company of Barbers guild, 11

Wrapping/rodding techniques, 594

Wrinkle treatment, 199

Wrinkles, 321

Written agreements, documents that govern the opening of a barbershop, including leases, vendor contracts, and employee contracts; all of these detail, usually for legal purposes, who does what and what is given in return, 758–759

X

X-rays, 214

Y

Yelp, 751

Z

0 elevation, 398

Zygomatic bones, bones that form the prominence of the cheeks, 150

Zygomatic nerve, affects the muscle of the upper part of the cheek, 159, 160

Zygomaticus major muscles, muscle extending from the zygomatic bone to the angle of the mouth; pull the mouth upward and backward as in laughing, 155

Zygomaticus minor muscles, muscles on both sides of the face that extend from the zygomatic bone to the upper lips; pull the upper lip backward, upward, and outward, as when you are smiling, 155

